

Bid Timing and Market Efficiency in the Energy Transition

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Abstract

Utility-scale batteries play an increasingly active role in electricity markets by arbitraging energy across time, yet their efficiency depends critically on market design. This paper studies a prevalent feature of real-time electricity auctions: bid lead time, which requires batteries to commit bids in advance of dispatch. Using comprehensive 15-minute bidding and settlement data for all utility-scale batteries in ERCOT from 2018–2025, we document active dynamic bidding, rapid responses to price shocks, and the importance of recent price information for forecasting. We develop an equilibrium model of electricity markets with dynamic battery bidding and show that longer bid lead times substantially reduce operational efficiency and profits, and dampen complementarity between batteries and renewable energy. Allowing bids to depend on state of charge significantly mitigates these losses. Our results highlight bid lead time as a key determinant of battery performance and market outcomes.

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1 Introduction

Utility-scale batteries have expanded rapidly in capacity in the United States and play a critical role in the energy transition. In wholesale electricity markets, batteries serve dual functions: they store energy to complement intermittent renewable generation and act as energy arbitrageurs that smooth intra-day price fluctuations. Batteries actively participate in real-time energy auctions, charging when prices are low and discharging when prices are high. The efficiency of these bidding strategies depends critically on the information available to forecast future prices and optimally time charging and discharging decisions. Despite growing economic interests in energy storage, limited attention has been paid to how batteries operate in electricity markets, how efficiently they bid as energy arbitrageurs, and how bidding (in)efficiencies affect market equilibrium and the broader energy transition.

This paper investigates a key market design feature of real-time electricity auctions that is particularly relevant to battery bidding: the bid lead time, defined as how far advanced batteries should commit their bids. The bid lead time is a common design in the wholesale electricity market around the world and varies across markets. For example, in California ISO (CAISO), batteries need to submit bids for every 60-minute period, commit 75 minutes before start of period, whereas in MidContinent Independent System Operator (MISO), Southwest Power Pool (SPP), and ISO New England (ISO-NE), batteries need to commit their bids 30 minutes in advance. Bid lead time has also received regulatory attention. The CAISO Market Surveillance Committee, for example, has noted that “CAISO should implement rules that enable battery operators to more efficiently manage their state of charge over the day, such as more frequent offer price or limit changes with shorter time lags, state of charge dependent bids and offers, and/or perhaps other design changes.” The bid lead time affects the efficiency of battery operations, because it limits access to the most recent price information and constrains the flexibility of adjusting the state of charge.

We evaluate how bid lead time shapes battery bidding efficiency in the context of Electric Reliability Council of Texas (ERCOT), where rapidly expanding storage capacity and detailed high-frequency bidding data provides an ideal setting to study battery bidding behavior and real-time market designs. Moreover, ERCOT effectively has no meaningful bid lead time in real-time, which provides an ideal setting where batteries update bids using near real-time information.

We compile a comprehensive data set covering all utility-scale batteries in ERCOT from 2018 to 2025, and we access the real-time bidding and settlement information for every 15 minutes. Using these high-frequency data, we document three key data patterns. First, we find that batteries are actively bidding into the real-time energy market, charging during

low-price periods and discharging during high-price periods to maximize profits. Batteries submit high generation bid prices to avoid being dispatched too early and to preserve the high option value of stored energy. As prices approach their peak, batteries sharply lower generation bid prices to ensure dispatch and capture high market prices. Second, we find that batteries respond to unexpected market shocks by quickly adjusting the generation bid prices. This behavior demonstrates that batteries actively exploit the real-time information and flexibly adjust their bidding strategy to accommodate the unexpected changes in market conditions. Third, since accurate prediction of future prices is a key prerequisite for batteries to earn profits, we demonstrate a critical role of recent price information in reducing forecast errors.

Motivated by these empirical facts, we develop an equilibrium model of electricity markets, which embeds a dynamic model of battery bidding. Batteries participate in the real-time market by submitting charge and discharge bids every 15 minutes, which determine whether they demand or supply energy depending on realized prices. Because electricity is storable, batteries face an inherently dynamic problem: discharging this period generates immediate profits but reduces the ability to supply energy in future high-price periods, while charging stores energy for future arbitrage. We model this trade-off as a dynamic optimization problem in which batteries choose bids to maximize expected profits, taking into account stochastic electricity prices and the evolution of their state of charge. A key feature of the model is that bidding behavior encodes the inter-temporal opportunity cost of stored energy. Optimal discharge bids reflect both current price conditions and the expected value of retaining energy for future periods, while optimal charge bids reflect the discounted value of increasing state of charge. In the baseline environment, bids are submitted just before dispatch, allowing batteries to condition on up-to-date information about prices and system conditions. This benchmark corresponds to a frictionless setting in which batteries can respond optimally to real-time shocks.

We then introduce bid lead time as a market design constraint such that bids for a given interval must be submitted several periods in advance. This modification generates three key frictions. First, batteries must form expectations using outdated price information, which increases forecast uncertainty. Second, they no longer observe their actual state of charge at the time of dispatch, but instead must plan based on its expected distribution. Third, they lose the ability to adjust bids in response to newly realized shocks. As a result, bidding becomes a problem of decision-making under imperfect information and commitment, leading to inefficient inter-temporal allocation of energy. We embed this dynamic bidding problem in a market equilibrium framework in which battery dispatch interacts with aggregate supply and demand to determine electricity prices. Using the model, we quantify how bid lead time

distorts bidding behavior, reduces arbitrage efficiency, and lowers battery profits. These inefficiencies propagate to the market level, affecting price dynamics, consumer surplus, and the distribution of rents across market participants.

We estimate the model using high-frequency data from ERCOT. We first recover a semi-parametric price process that preserves intraday dynamics and price spikes, and then estimate battery-specific cycling costs using a simulated method of moments, matching observed charging and discharging behavior. The estimated cycling costs range from roughly \$3 to \$30 per MWh for most batteries, consistent with engineering benchmarks, and are tightly disciplined by observed cycling frequency and price volatility. The model fits key features of battery operations, including intraday charging and discharging patterns.

Using the estimated model, we quantify the impact of bid lead time on behavior and outcomes. Longer lead times lead batteries to submit more conservative and less targeted bids, causing them to discharge too early and miss peak-price periods. This distortion generates substantial profit losses: relative to last-minute bidding, a 30-minute lead time reduces profits by about 17%, while a 75-minute lead time reduces profits by roughly 25%. These effects are amplified by higher price volatility and mitigated by longer-duration batteries, which have greater inter-temporal flexibility.

Next, we explore a commonly discussed regulatory proposal of state-of-charge-dependent bids such that batteries could submit a contingent bid price dependent on the realized state of charge. Due to computation limitations, we evaluate a parsimonious two-point state-of-charge-dependent bids. We find that state-of-charge-dependent bids brings substantial improvement to batteries' operation profits, which mitigate quantity uncertainty and allow contingent plans based on different realizations of state of charge. Under 30-minute bid lead time, state-of-charge-dependent bids improve the average profit by 12.3%, and under 75-minute bid lead time, state-of-charge-dependent bids improve the average profit by 15.8%. Allowing contingent bidding plans significantly lowers the uncertainty towards quantities, and improves batteries' operation profits.

Finally, we embed the model in a market equilibrium framework to assess aggregate impacts. Removing bid lead time lowers average electricity prices, especially in the upper tail, by enabling batteries to more effectively arbitrage peak demand. The resulting gains accrue primarily to consumers and battery operators, while reducing profits for conventional generators. Overall, we find that bid timing is a quantitatively important dimension of market design: even relatively short lead times generate meaningful inefficiencies in storage operations and distort equilibrium outcomes in electricity markets.

This paper contributes to the following three strands of literature. First, this is the first economics paper, to our knowledge, to empirically evaluate batteries dynamic bidding be-

havior, which play a critical role in the equilibrium on the wholesale electricity market. Our paper moves away from common assumptions that batteries are perfect energy arbitrageurs and builds in inefficiency from the regulatory environment such that their operation might be compromised. Our paper thus complements previous work on the economic values of energy storage, including [Butters et al. \(2025\)](#), [Holland et al. \(2024\)](#), [Karaduman \(2021\)](#), and [Lamp and Samano \(2022\)](#). Previous work has also examined various aspects of battery operations, exploring the market power ([Andrés-Cerezo and Fabra, 2023](#)), complementarity with renewable energy ([Andrés-Cerezo and Fabra, 2025](#)), congestion benefits ([Kirkpatrick, 2020](#)), ancillary services ([Reynolds, 2025](#)), and long run equilibrium under renewable portfolio standards ([Palley et al., 2024](#)). Our paper exploits the high-frequency dispatching and bidding data and estimates fundamental primitives of battery operations.

Second, this paper contributes to a large literature on the design and performance of wholesale electricity markets. Foundational empirical work has documented inefficiencies and market power in restructured electricity markets ([Borenstein et al., 2002](#); [Bushnell et al., 2008](#); [Mansur, 2008](#)), the gains from competitive dispatch relative to centralized regulation ([Cicala, 2022, 2015](#)), and the role of specific market design features such as complementary bidding and startup costs ([Reguant, 2014](#)), emissions cost pass-through ([Fabra and Reguant, 2014](#)), and renewable integration ([Wolak, 2003](#)). Within this literature, previous papers have examined strategic bidding ([Hortaçsu and Puller, 2008](#)), sequential markets ([Ito and Reguant, 2016](#)), real-time pricing ([Allcott, 2012](#)), bidding sophistication ([Hortaçsu et al., 2019](#)), and forward commodity markets ([Jha and Wolak, 2023](#)). This paper incorporates the bidding behavior of energy storage, highlights its unique dynamic feature, and evaluates the consequences of a common but understudied market design feature — bid lead time — along with its potential remedies.

Third, and more broadly, this paper relates to the broader literature on auction design, information structure, and dynamic efficiency, where equilibrium outcomes depend critically on the timing and precision of the information available to bidders. Electricity markets are large-scale, repeated, multi-unit divisible-good auctions, a setting whose theoretical foundations are developed in [Wilson \(1979\)](#), [Back and Zender \(1993\)](#), [Ausubel et al. \(2014\)](#), and surveyed by [Milgrom \(2004\)](#); [Klemperer \(2004\)](#). A growing empirical literature exploits bid-level data to recover preferences and welfare in divisible-good auctions, including studies of treasury and central-bank auctions ([Hortaçsu and McAdams, 2010](#); [Kastl, 2011](#); [Hortaçsu and Kastl, 2012](#)), as well as broader empirical auction settings ([Athey et al., 2011](#); [Hortaçsu and Perrigne, 2021](#)). A second related strand studies how information frictions — asymmetric private information, imperfect signals, and learning — shape auction outcomes and the surplus of bidders versus auctioneers ([Milgrom and Weber, 1982](#); [Bergemann and](#)

[Morris, 2013](#)). The closest related work studies how auction timing and dynamic participation affect efficiency ([Chaves and Ichihashi, 2024](#); [Hendricks et al., 2025](#)). In contrast, we study a different margin of timing in a dynamic auction: a commitment constraint on *when* bids must be submitted, which restricts the information set available to bidders at the moment of decision-making. Because batteries’ inter-temporal opportunity cost of energy evolves stochastically, such commitment generates dynamic inefficiencies even in otherwise well-functioning markets and even absent market power or private information. Our analysis thus connects a classical auction-design margin: the information available to bidders at the time of bid submission, to a quantitatively important feature of modern electricity market design.

2 Background and Data

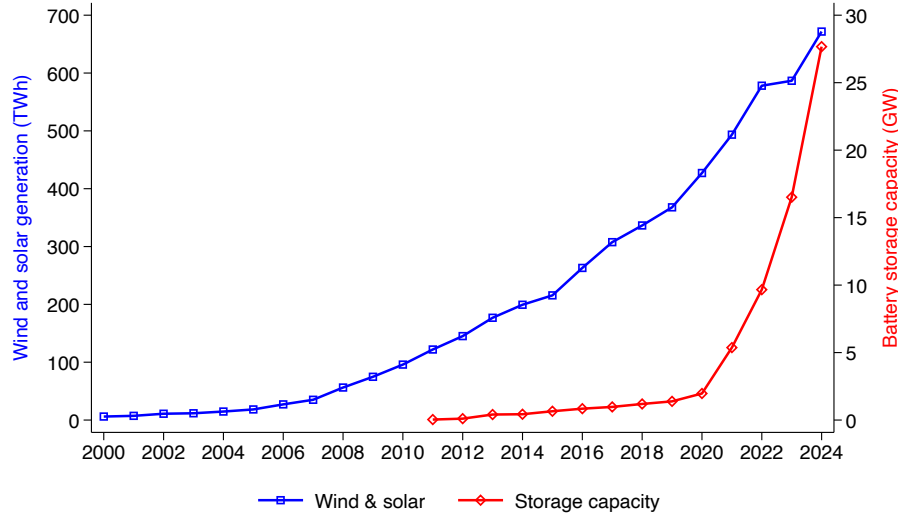
2.1 Energy Storages in Energy Transition

Wholesale electricity markets in the United States and abroad are undergoing a structural transformation driven by the rapid growth of intermittent renewable generation. As shown in Figure 1, wind and solar accounted for a negligible share of U.S. electricity generation in the early 2000s but exceeded 15 percent in 2024. This shift fundamentally changes the supply side of electricity markets. While fossil-fuel generation depends on fuel input and operating decisions, wind and solar output depends primarily on weather conditions and has near-zero marginal operating cost. As renewable output rises and falls, the demand left for conventional generators becomes more volatile and harder to predict. Similar changes are evident across major U.S. wholesale electricity markets, as well as in international markets with high renewable penetration, including the Australian National Electricity Market and several European power systems. A direct consequence of higher renewable shares is a sharp increase in short-run real-time price volatility. With substantial intermittent electricity supply, real-time prices spike during periods of low renewable output or unexpected demand, and fall to zero or below during periods of abundant wind and solar generation. Higher real-time volatility raises the option value of flexibility and the returns to intra-day arbitrage, creating a strong economic rationale for the rapid expansion of energy storage.

In response, utility-scale battery capacity expanded modestly after 2010 and experienced rapid growth after 2020. Total US battery capacity has exceeded 27 GW in 2024, a thirteen-fold increase from roughly 2 GW in 2020. Batteries are emerging as the principal new source of intra-day flexibility on the grid, and they play dual roles in wholesale electricity markets.

First, they store energy and complement intermittent renewable generation. Because wind and solar output often peak at times that do not coincide with peak electricity demand, batteries can shift energy across time by charging during periods of high renewable generation and discharging during periods of high demand. Second, batteries act as energy arbitrageurs. By charging when prices are low and discharging when prices are high, batteries smooth intra-day price fluctuations. Their economic value depends critically on how effectively they can translate volatile real-time price signals into well-timed charging and discharging decisions.

Figure 1: Market Evolution of Renewable Generation and Battery Capacity



Notes: This figure plots the time trend of wind and solar generation (in blue, y-axis on the left) and the battery capacity (in red, y-axis on the right) in the US from 2000 to 2024.

Our paper focuses on Electric Reliability Council of Texas (ERCOT) which operates the wholesale electricity market in Texas. Texas now hosts the second largest battery capacity in the US, following California. Unlike California, where batteries are often co-located with solar farms and are predominantly four-hour systems, batteries in Texas are mostly stand-alone facilities and are primarily one-hour or two-hour batteries, as shown in Appendix Figures A1 and A2. Batteries in ERCOT participate in both the day-ahead market, primarily to provide ancillary services, and the real-time market for energy arbitrage.¹

¹Ancillary services are purchased by ERCOT from the day-ahead market to balance electricity supply and demand next day on the grid and mitigate real time operational issues. Ancillary services in ERCOT include different types: regulation service (regulation up and down), responsive reserve service, non-spin reserve service, and ERCOT contingency reserve service (ECRS). Different types of ancillary services vary across three core dimensions: response speed, direction (inject power or absorb power), and duration. Energy storage participates in all kinds of ancillary services.

Batteries in ERCOT actively participate in the real-time wholesale auctions every five minutes, with settlements occurring every fifteen minutes. In each interval, a battery submits a *generation* bid to supply energy and a *load* bid to demand energy. These bids are interval- and unit-specific step functions. We illustrate the shape of bid functions for one of the largest batteries in ERCOT, Turquoise Storage, on one hot summer day in 2024, as shown in Appendix Figures A3 and A4. Generation bid prices indicate the minimum price that batteries are willingness to supply a certain amount of energy. They are capped from above at \$5,000/MWh, and a generation bid price at the cap could effectively avoid dispatch. Load bid prices indicate the maximum price that batteries are willingness to accept to demand a certain amount of energy. They are capped from below at -\$250/MWh, and a load bid price at the floor could effectively avoid charging.

To fully exploit these arbitrage opportunities, batteries must forecast future prices and dynamically decide when to charge and when to discharge at high frequency. Within ERCOT, Appendix Figure A5 shows that the dispersion between real-time and day-ahead settlement point prices (DART spread), a direct measure of price volatility realized after day-ahead clearing, has widened substantially over our sample period. Since day-ahead prices are known to batteries at the time of real-time bidding, a wider DART spread directly reflects growing uncertainty about real-time prices conditional on all information available before the operating day, making accurate intra-day forecasting central to battery profitability.

2.2 Bid Lead Times in Wholesale Electricity Markets (ISO/RTOs)

The bidding efficiency of batteries depends crucially on the design of the real-time energy auctions, and the wholesale electricity market in general. Obstacles embedded in the market design could potentially hinder efficient bidding of batteries, dampen their operation profitability, and shape the relationship between energy storage and renewable sources. We focus on one important market design feature: the bid lead time, defined as how far advanced batteries should commit their bids. The bid lead time is a common design in the wholesale electricity market around the world, and it varies across markets. For example, in CAISO, storage resources must submit bids up to 75 minutes prior to the operating hour, after which bids remain fixed for the duration of the hour, whereas in MISO, SPP, and ISO New England, batteries need to commit their bids 30 minutes in advance. In comparison, ERCOT represents a market design with near-zero bid lead time. ERCOT’s real-time market features a 5-minute dispatch interval and allows market participants to update energy offer curves at high frequency. Appendix Table A1 summarizes real-time bid lead times and bid

commitment lengths across the major US ISOs/RTOs and several international markets.²

As noted by a recent report on energy storage deployment ([GridLab, 2026](#)), long bid lead time is one of the major challenges on the energy market for batteries. Since most wholesale electricity markets require participants to submit bids 30 to 75 minutes prior to the operating hour, while market dispatch and settlement occur at much finer temporal resolution (e.g., five-minute intervals), this institutional feature creates a fundamental timing mismatch to batteries whose marginal costs are inherently dynamic. Because the true value of stored energy evolves continuously while bids remain fixed, battery operators must commit to bids that may quickly become outdated, leading to inefficient dispatch and distorted incentives. As emphasized by the report, “this contrast highlights that bid lead times and lock-in requirements are not merely operational details but central design features that shape the efficiency and economic performance of storage participation in wholesale markets.”

Bid lead time has also received regulatory attention. The CAISO Market Surveillance Committee has noted that “CAISO should implement rules that enable battery operators to more efficiently manage their state of charge over the day, such as more frequent offer price or limit changes with shorter time lags, state of charge dependent bids and offers, and/or perhaps other design changes.” As emphasized by the Department of Market Monitoring of CAISO, these bids are formed based on expectations of the resource’s current and future state of charge, yet actual dispatch occurs after this submission, potentially leading to substantial deviations between expected and realized state of charge. Because of this timing mismatch, storage operators must commit to opportunity-cost-based bids under incomplete information, while their physical state continues to evolve between bid submission and dispatch. Stakeholder comments reinforce the implications of this design. Terra-Gen, one of the major energy storage owners and operators, highlighted that storage operators cannot adjust bids closer to real time, even though realized market outcomes and thus state of charge availability depend on uncertain dispatch conditions beyond the operator’s control.³ This can result in outcomes that are not fully aligned with the batteries’ actual operating constraints, motivating the need for uplift payments to compensate for inefficient dispatch outcomes induced by market design limitations.

These concerns are echoed in our own conversations with battery operators, traders, and market analysts active in multiple ISOs. Across these conversations, bid lead time was consistently flagged as the single most impactful real-time market design feature affecting

²Prior to recent reforms, offers were effectively locked several minutes before dispatch through a short adjustment period, implying a small but non-zero bid lead time. However, with the implementation of NPRR1058 in 2024, ERCOT allows offer updates during the operating period, effectively eliminating binding bid timing constraints.

³This is quoted from the [CAISO shareholder meeting](#).

storage operations in markets other than ERCOT, and as the leading source of operational friction that participants would prioritize for reform.

Taken together, the bid lead time as a common market design feature across many ISO/RTOs, imposes a commitment constraint with information lag, where storage resources must make forward-looking bidding decisions under uncertainty while facing limited ability to adjust to realized conditions. This generates a wedge between bid-based optimization and physical feasibility which could have important implications for market efficiency.

2.3 Data

We compile a comprehensive dataset covering all 338 utility-scale batteries (Energy Storage Resources) registered in ERCOT from 2018 to 2025.⁴ We obtain bidding and settlement data from ERCOT’s publicly available market data, which covers both the day-ahead market, including ancillary services, and the real-time market.⁵ For each battery in each 15-minute market interval, we observe submitted bids, bid awards, and settlement point prices.⁶ Prices in ERCOT are location-specific: each battery is assigned to a settlement point, and we match all observations to the corresponding real-time settlement point price.

Batteries participate as two distinct resource types: as a generation resource when discharging, and as a load resource when charging. Each resource submits a separate step-function offer curve per 15-minute delivery interval. The generation offer curve consists of up to 35 price-MW pairs, $\{(q_k^g, p_k^g)\}_{k=1}^K$, where q_k^g is the offered level of output (in MW) when the settlement point price falls in the range $[p_k^g, p_{k+1}^g)$. The load bid curve contains up to 10 price-MW pairs, $\{(q_k^l, p_k^l)\}_{k=1}^K$, where q_k^l is the bid level of consumption (in MW) when the settlement point price falls in the range $(p_{k-1}^l, p_k^l]$.⁷ To summarize bidding behavior into a scalar measure, which we use in our descriptive evidence below, we compute a MW-weighted average bid price for each resource-interval observation. For the generation offer, the average

⁴Our sample ends on December 4, 2025 because ERCOT’s Real-Time Co-Optimization + Batteries (RTC+B) market redesign took effect on December 5, 2025, fundamentally altering how batteries bid and settle in the market.

⁵The data is mainly compiled from 60-Day Security Constrained Economic Dispatch (SCED) Disclosure files, published on a rolling 60-day delay following each delivery date.

⁶In ERCOT’s real-time market, the dispatch algorithm (Security-Constrained Economic Dispatch, or SCED) runs every five minutes, producing locational marginal prices and dispatch instructions at the 5-minute frequency. Energy offer and load bid curves are submitted for 15-minute delivery intervals, and the real-time settlement point price used for billing each 15-minute interval is the time-weighted average of the three 5-minute prices within the interval. Throughout the paper we therefore treat the 15-minute interval as the unit of bidding while preserving the 5-minute dispatch resolution when modeling price formation.

⁷The ceiling of generation bid price p_K^g is set to the system-wide cap of \$5,000/MWh. The floor of load bid price p_0^l is set to the floor of -\$250/MWh and the ceiling of load bid price p_K^l is set to the system-wide cap of \$5,000/MWh. These bounds are defined by ERCOT’s market rules and apply to all resource types.

price is

$$\bar{p}^g = \frac{\sum_k p_k^g \cdot (q_k^g - q_{k-1}^g)}{\sum_k (q_k^g - q_{k-1}^g)},$$

with the load bid average computed analogously. This measure weights each price step by its MW coverage, capturing the price of a representative megawatt across the full offer curve.⁸

ERCOT’s day-ahead market differentiates between two types of participation with distinct implications for real-time operations. For energy offers, the day-ahead market functions as a financial-forward market: a battery awarded energy in the day-ahead market receives the day-ahead settlement point price for the awarded quantity but incurs no physical dispatch obligation; actual dispatch is determined independently by the real-time market based on the battery’s real-time offer curves, with any imbalance between day-ahead-awarded and actual real-time quantities settled at the real-time price. Because day-ahead energy participation is purely financial and does not constrain real-time behavior, we make limited use of it in the analysis of this paper.

By contrast, ancillary service awards from the day-ahead market operate differently and are physically binding. A battery awarded, for instance, Regulation Up capacity in the day-ahead market must hold that capacity available for deployment in real-time, directly constraining the energy available for arbitrage. We take ancillary service participation into account when characterizing the data, but our main analysis focuses on batteries that participate exclusively in the real-time energy market.

We complement the bidding data with battery characteristics from the U.S. Energy Information Administration (EIA) Form EIA-860. This annual survey provides plant-level information on installed power capacity (MW), energy capacity (MWh), commercial operation date, and geographic location.⁹ We match ERCOT resource names to EIA plant identifiers using a combination of text-based name matching and manual verification. These data allow us to characterize the variation in battery size and duration within our sample, and to normalize bid quantities by capacity where needed.

⁸A practical challenge is that batteries routinely submit extreme prices that reflect dispatch preferences rather than economic valuations. On the generation side, prices near the \$5,000/MWh cap signal a preference not to discharge; on the load side, prices near the -\$250/MWh floor signal a preference not to charge, while prices near the ceiling signal unconditional willingness to charge. We therefore apply two cleaning steps before computing averages. For generation offers, we truncate prices above \$1,000/MWh to missing, treating bids above this threshold as dispatch-avoidance signals, and winsorize prices below -\$100/MWh to -\$100, preserving these as genuine willingness-to-dispatch signals while preventing their scale from distorting the average. For load bids, the logic is symmetric: prices below -\$100/MWh are truncated to missing (charging-avoidance signals), and prices above \$1,000/MWh are winsorized to \$1,000.

⁹Energy capacity (MWh) is installed power capacity (MW) multiplied by battery duration.

3 Descriptive Evidence

Fact 1: Batteries bid dynamically and option value matters

We document dynamic battery bidding patterns in Figure 2. For each 15-minute interval of the day, we compute average generation and load bid prices. ERCOT real-time prices exhibit a clear bimodal pattern, with a modest morning peak around 7 a.m. and a much larger evening peak around 7 p.m., both coinciding with periods of high electricity demand. Prices are lowest from the morning through early afternoon (approximately 8 a.m. to 2 p.m.), reflecting the abundance of renewable generation. In response to this intraday price variation, batteries adjust both their generation and load bid prices dynamically. Generation bid prices reflect batteries’ willingness to sell energy, and higher generation bid prices reduce the likelihood of dispatch. Conversely, load bid prices capture batteries’ willingness to buy energy and higher load bid prices increase the likelihood of being charged.

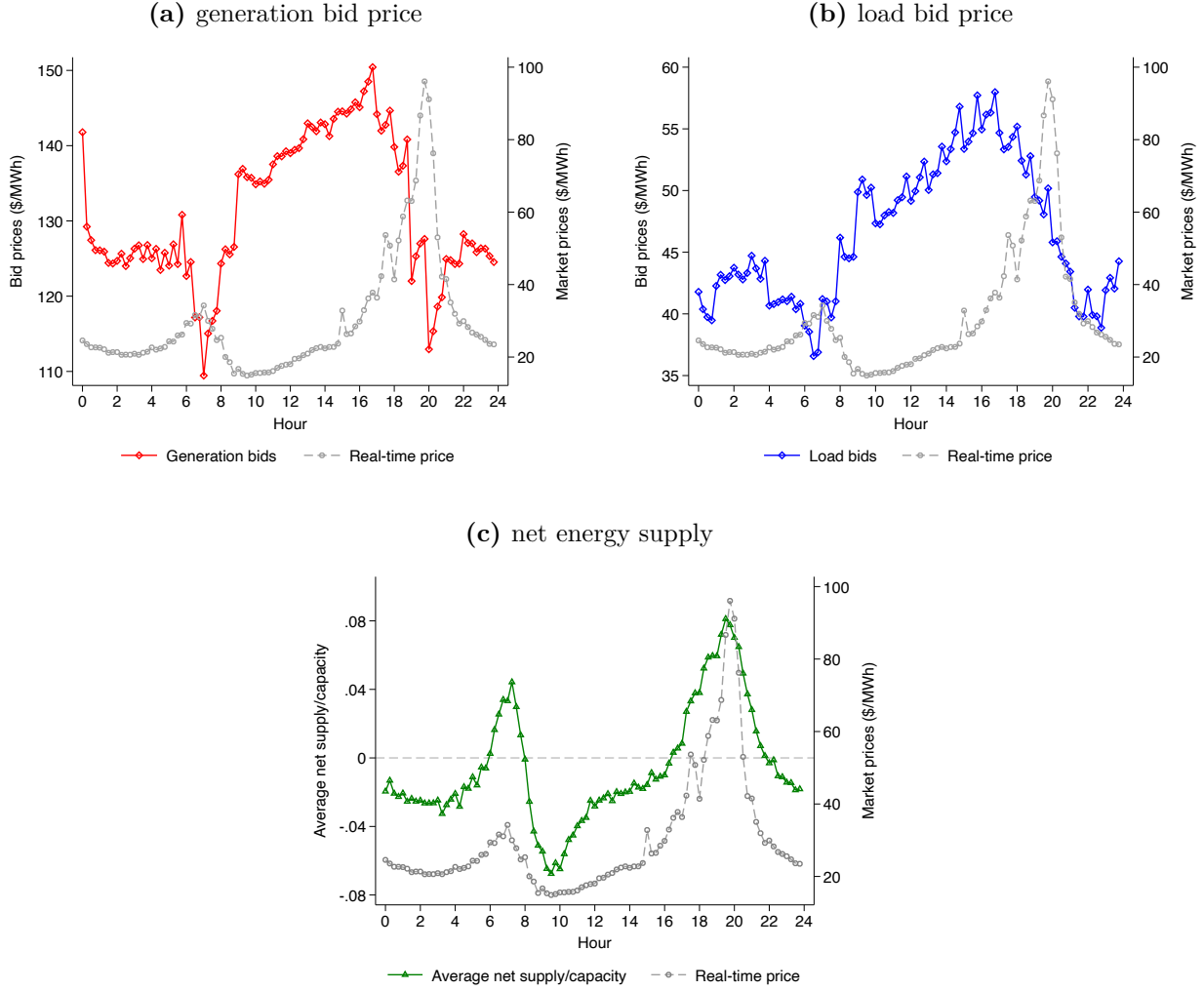
As shown in Panel (a), batteries submit relatively high generation bid prices several hours before the peak period to avoid being discharged too early, reflecting the high option value of stored energy. It is more profitable to retain energy for later, higher-price hours than to sell during periods of relatively low prices. As the price peak approaches, generation bid prices drop sharply to ensure dispatch, allowing batteries to sell when prices are highest. We observe a similar pattern around the morning peak, although the magnitude is more modest.

As shown in Panel (b), we find that load bid prices exhibit a comparable intraday pattern, albeit at lower levels. Batteries submit relatively high load bid prices several hours before the peak period, again reflecting the high option value of stored energy and the incentive to secure full charge ahead of high-price periods. As peak hours arrive, load bid prices decline sharply, as batteries avoid charging when grid prices are elevated. As with generation bids, the same pattern for load bid prices appears around the morning peak, though with smaller magnitudes.

Consistent with the bidding strategy, the average net supply, defined as the difference between the total discharged and charged energy, closely track the price pattern, as shown in Panel (c). Batteries are net energy suppliers during morning and evening peak hours, and net energy consumers during the rest of the day. Overall, these patterns reveal a clear dynamic bidding behavior of batteries. They are actively bidding into the real-time energy market to charge during low-price periods and discharge during high-price periods to maximize profits. The intra-day bid price patterns follow closely the option values of stored energy.¹⁰

¹⁰Most batteries in ERCOT have adopted automated bidding tools to allow real-time responses of bid functions to market information and forecast. See early report on automated bidding by [Utility Drive](#) and a [report](#) from Key Capture, one of the largest battery owners in ERCOT. We discuss one concrete example of automated bidding adoption in Appendix B. We find that most recently installed batteries have not engaged

Figure 2: Bid Prices and Net Energy Supply of Batteries



Notes: This figure plots the average generation bid prices (red line in Panel (a)), load bid prices (blue line in Panel (b)), and the net energy supply as a share of capacity (green line in Panel (c)) in 2024. We truncate both generation and load bid prices at an upper bound of \$1,000/MWh and winsorize them at a lower bound of -\$100/MWh. Real-time market prices are plotted in grey with y-axis on the right for each panel.

Fact 2: Batteries respond to unexpected market shocks

We examine how batteries respond to unexpected market shocks. We identify shocks at the each battery's grid location (its settlement point, the node at which its dispatch is in manual bidding since coming online.

priced) and at 15-minute-interval level by imposing two restrictions.¹¹ First, shocks must be associated with unplanned generation or transmission outages lasting more than two hours. Second, we require that these outages generate abrupt and substantial changes in market conditions. Specifically, we classify an interval as incurring a shock if, following the onset of the outage, the real-time price exceeds the day-ahead price by at least \$100/MWh, while the real-time-day-ahead price spread remains within $[-50, 50]$ \$/MWh over the preceding three hours. This criterion isolates episodes in which real-time prices deviate sharply from day-ahead expectations despite relatively stable conditions immediately beforehand, capturing genuinely unanticipated disruptions due to outages rather than gradual market adjustments.

We estimate the following event-study specification to evaluate how batteries respond to unexpected market shocks.

$$y_{it} = \sum_{k=-12, k \neq -1}^{12} \beta_k \mathbb{1}(T_{it} = k) + \alpha Price_{it}^{DAM} + X_{it} + \epsilon_{it}. \quad (1)$$

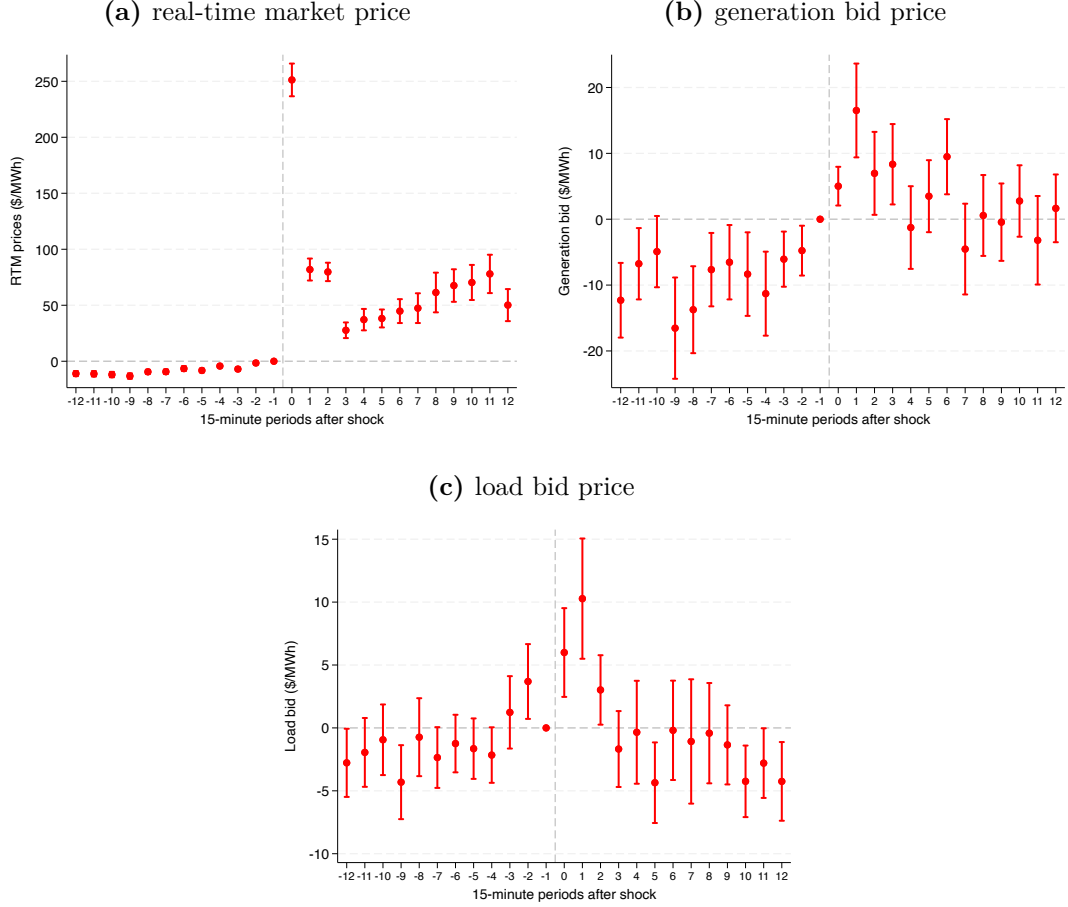
We index battery by i and 15-minute intervals by t . The dependent variable y_{it} denotes key outcome variables, including market prices and bid prices. T_{it} represents event time relative to the start of a market shock, with $k = -1$ omitted as the reference period. Our key coefficients of interest is β_k , which measures how unexpected market shocks affect the dependent variable k intervals relative to the onset of the shock. We control for day-ahead market prices $Price_{it}^{DAM}$, and include a rich set of fixed effects X_{it} , consisting of battery fixed effects interacted with month and hour dummies. These fixed effects absorb battery characteristics, seasonality in electricity demand, intra-day volatility, as well as variations in generation resource mixes around each battery. The key identification assumption is that, conditional on the rich controls of day-ahead market prices and unobserved shocks at the battery, month, and hour levels, the timing of market shocks is exogenous to individual batteries.

The event-study estimation results are presented in Figure 3. We confirm that our identified unexpected market shocks drive price spikes in Panel (a), as we observe an abrupt rise in real-time market price at the onset of the shock, and the increase in the price level lasts for more than three hours. On the contrary, we observe a stable level of the real-time market price before the event. More importantly, we find that batteries respond to such shocks by increasing the generation and load bid price levels. Batteries may anticipate persistence in price shocks and therefore optimally respond by increasing both generation and load bids, allowing them to cycle multiple times during periods of elevated prices. This

¹¹A settlement point is defined as the location where prices are calculated for settlement, including Resource Nodes, Load Zones, and Hubs, according to [official protocols](#).

behavior demonstrates that batteries actively exploit the real-time information and flexibly adjust their bidding strategy to accommodate the unexpected changes in market conditions.

Figure 3: The Impacts of Unexpected Market Shocks



Notes: This figure plots the estimation results of equation (1). Panel (a) plots the impacts of unexpected market shocks on real-time prices, panel (b) plots the coefficients on generation bid prices, and panel (c) plots the coefficients on load bid prices. 95% confidence intervals are constructed from clustered standard errors at the battery level.

Fact 3: Recent prices are most informative for price projections

We investigate the role of recent price information in shaping expectations about future prices. Using real-time market prices at a 5-minute frequency, we estimate the following empirical specification to evaluate how much variation in current prices can be explained by past price realizations.

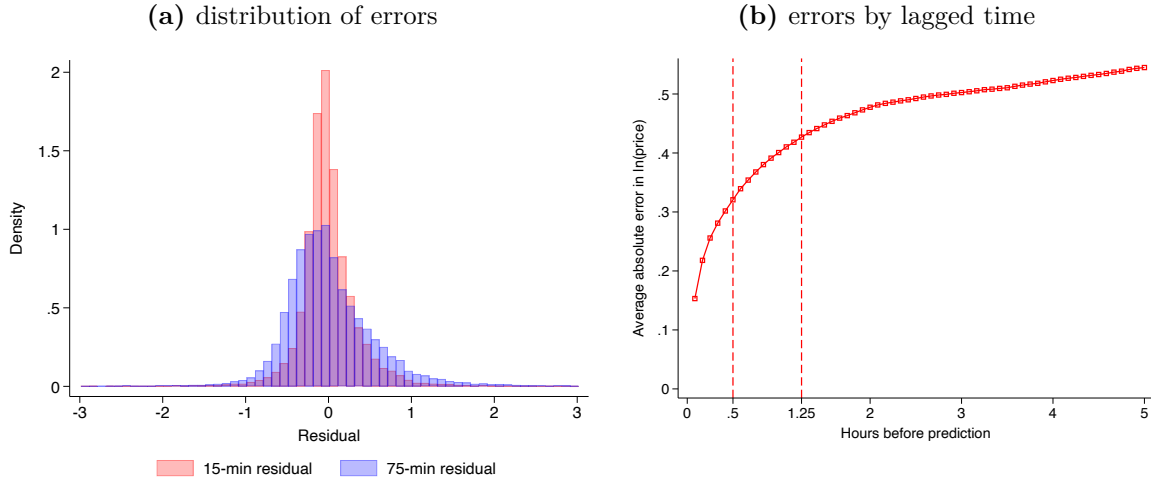
$$\log p_{d,h} = \gamma_h \log p_{d,h-\tau} + \phi \text{Price}_{d,h}^{DAM} + X_{d,h} + \epsilon_{d,h}. \quad (2)$$

Here, d denotes the day and h indexes the intra-day 5-minute interval. The dependent variable is the log real-time market price at day d and interval h , and the key independent variable is lagged log real-time prices τ intervals earlier. We allow the coefficients on lagged prices to vary flexibly across intra-day intervals. Moreover, we control for hourly day-ahead market prices $\text{Price}_{d,h}^{DAM}$, and include month fixed effects to absorb seasonal variations in prices.

The main results are summarized in Figure 4. Panel (a) plots the distribution of prediction residuals $\hat{\epsilon}_{d,h}$ under two specifications: one using only a 15-minute lag ($\log p_{d,h-3}$) to form a 15-minute-ahead forecast, and another using only a 75-minute lag ($\log p_{d,h-15}$) to form a 75-minute-ahead forecast. Extending the forecasting horizon from 15 to 75 minutes substantially increases the dispersion of prediction errors, with the standard deviation rising by 43.9%.

Panel (b) plots the average absolute prediction errors as a function of forecasting horizon. We observe a sharp increase in errors, particularly within the first two hours. Overall, these results highlight the importance of recent price information for accurately predicting future real-time prices.

Figure 4: Projection Errors of Prices



Notes: This figure plots the estimation results of equation (2). Panel (a) plots the distribution of forecast errors when $\tau = 3$ (a 15-minute-ahead prediction) and when $\tau = 15$ (a 75-minute-ahead prediction), while panel (b) plots the average absolute prediction errors as a function of forecasting horizon.

Taking stock, these empirical patterns document batteries' dynamic bidding incentives and illustrate how batteries adjust bid prices to exploit intra-day price volatility to maximize profits. Moreover, we also show that batteries respond rapidly to unexpected market shocks in the real-time market. Accurate prediction of future prices is a key prerequisite for earning profits, and we document the critical role of recent price information in reducing forecast errors. These findings motivate our structural model, in which we characterize batteries' dynamic bidding behavior in the real-time market, and analyze how batteries optimally adjust their bids in the presence of bid lead time that restricts the effective prediction horizon.

4 Structural Model

This section develops a structural model that links battery bids, state-of-charge dynamics, bid lead time, and market-clearing prices. We first characterize optimal last-minute bids, then introduce lead-time constraints, extend the model to state-of-charge-dependent bids, and finally embed battery bidding in market equilibrium.

4.1 Dynamic Bidding of Batteries

We index batteries by i , each with energy capacity k_i , the maximum amount battery i can discharge. Time t is discretized into 15-minute intervals. We index days by d and intra-day intervals by $h \in \{1, 2, 3, \dots, 96\}$. At the beginning of each interval t , battery i has a remaining state of charge $s_{it} \in [0, 1]$, defined as the level of charge relative to its capacity. We normalize energy quantities by k_i throughout, so bid quantities and profits are measured per unit of battery capacity. Batteries submit interval- and unit-specific step-function bids: a generation bid for discharging and a load bid for charging. For tractability, we approximate each step function with a single generation bid price b_{it}^g and a single load bid price b_{it}^l with exogenous bid quantities q_i^g and q_i^l for each interval. A battery is willing to supply q_i^g when the real-time market price exceeds b_{it}^g . Similarly, its load bid indicates willingness to charge when the real-time market price is below b_{it}^l .

We denote the real-time market price in interval h as p_h , which follows a Markov process $F_h(p_h|p_{h-1})$. Specifically, we assume that

$$p_t = \phi_h p_{t-1} + \xi_h + \epsilon_t. \quad (3)$$

Coefficients ϕ_h and ξ_h are the AR(1) slope and intercept for intra-day interval h . Henceforth we focus on within-day bidding and index time by h . We write $F_h(p_h)$ for the conditional

distribution $F_h(p_h|p_{h-1})$ when the conditioning information is clear.

Let $\delta < 1$ denote round-trip efficiency, defined as the ratio of energy discharged from the battery to the energy purchased for charging over a complete charge-discharge cycle. Since a battery cannot discharge below zero or charge above full capacity, feasible quantities are

$$q_h^g = \min\{q_i^g, s_h\}, \quad q_h^l = \min\left\{\frac{q_i^l}{\delta}, \frac{1-s_h}{\delta}\right\}.$$

Here q_h^l is the amount purchased from the grid, so charging increases state of charge by δq_h^l .

With $b_h^l \leq b_h^g$, the battery charges when $p_h \leq b_h^l$, discharges when $p_h \geq b_h^g$, and remains idle otherwise. The resulting net discharged energy a_h is

$$a_h = q_h^g \mathbb{1}(p_h \geq b_h^g) - q_h^l \mathbb{1}(p_h \leq b_h^l), \quad P_h(a_h) = \begin{cases} 1 - F_h(b_h^g), & a_h = q_h^g \\ F_h(b_h^l), & a_h = -q_h^l \\ F_h(b_h^g) - F_h(b_h^l), & a_h = 0. \end{cases}$$

Cycling costs are $\zeta|a_h|$, where ζ is the cost per unit of energy charged or discharged.¹² Expected profit equals discharge revenues minus charging costs and cycling costs:

$$\begin{aligned} \mathbb{E}_h[\pi(\mathcal{B}_h)] &= \mathbb{E}_h[a_h \times p_h - \zeta \times |a_h|] \\ &= \mathbb{E}_h\{[q_h^g \mathbb{1}(p_h \geq b_h^g) - q_h^l \mathbb{1}(p_h \leq b_h^l)] \times p_h - \zeta \times [q_h^g \mathbb{1}(p_h \geq b_h^g) + q_h^l \mathbb{1}(p_h \leq b_h^l)]\} \\ &= \underbrace{\int_{p_h \geq b_h^g} q_h^g p_h dF_h(p_h)}_{\text{revenues from discharging}} - \underbrace{\int_{p_h \leq b_h^l} q_h^l p_h dF_h(p_h)}_{\text{costs from charging}} - \underbrace{\zeta \times [q_h^g (1 - F_h(b_h^g)) + q_h^l F_h(b_h^l)]}_{\text{cycling cost}}. \quad (4) \end{aligned}$$

State of charge evolves according to

$$s_{h+1} = s_h - q_h^g \mathbb{1}(p_h \geq b_h^g) + \delta q_h^l \mathbb{1}(p_h \leq b_h^l), \quad P_h(s_{h+1}) = \begin{cases} 1 - F_h(b_h^g), & s_{h+1} = s_h - q_h^g \\ F_h(b_h^l), & s_{h+1} = s_h + \delta q_h^l \\ F_h(b_h^g) - F_h(b_h^l), & s_{h+1} = s_h. \end{cases}$$

Each interval, battery i chooses whether to charge, discharge, or remain idle via its submitted bids. By submitting a lower generation bid price in the current period, batteries are more likely to be dispatched and earn a profit from discharging energy. However, discharging lowers state of charge, reducing the ability to continue supplying energy in subsequent periods without recharging first. Conversely, when future prices are expected to be high,

¹²This functional form is a one-segment special case of [Xu et al. \(2017\)](#).

batteries tend to submit a high generation bid price to avoid being discharged early and raise their load bid price to recharge sooner. With discount factor β , the value function is

$$V_h(s_h, p_{h-1}) = \max_{b_h^l \leq b_h^g} \left\{ \underbrace{\mathbb{E}_h[\pi(b_h^g, b_h^l)]}_{\text{current-period profit}} + \underbrace{\beta \mathbb{E}_h[V_{h+1}(s_{h+1}, p_h) | s_h, p_{h-1}]}_{\text{continuation value}} \right\}. \quad (5)$$

The continuation value averages over the three possible state transitions induced by the bid prices:

$$\begin{aligned} \mathbb{E}_h[V_{h+1}(s_{h+1}, p_h) | s_h, p_{h-1}] = & \int_{p_h} \{ [1 - F_h(b_h^g)] V_{h+1}(s_h - q_h^g, p_h) \\ & + [F_h(b_h^g) - F_h(b_h^l)] V_{h+1}(s_h, p_h) \\ & + F_h(b_h^l) V_{h+1}(s_h + \delta q_h^l, p_h) \} dF_h(p_h | p_{h-1}). \end{aligned} \quad (6)$$

Assuming an interior optimum, the optimal bid prices satisfy

$$b_h^g = \zeta + \beta \times \frac{\int [V_{h+1}(s_h, p_h) - V_{h+1}(s_h - q_h^g, p_h)] dF_h(p_h | p_{h-1})}{q_h^g}. \quad (7)$$

$$b_h^l = -\zeta + \beta \times \frac{\int [V_{h+1}(s_h + \delta q_h^l, p_h) - V_{h+1}(s_h, p_h)] dF_h(p_h | p_{h-1})}{q_h^l}. \quad (8)$$

These expressions give the bid prices a direct interpretation. The generation bid price equals cycling cost plus the discounted opportunity cost of using stored energy now. The load bid price equals the discounted value of additional state of charge net of cycling cost. Higher cycling costs therefore make both bids more conservative and reduce expected cycling.

4.2 Bidding under Lead Time

Under bid lead time, bids for interval h must be submitted z periods earlier, at the beginning of interval $h - z$. Relative to last-minute bidding, lead time changes the battery's dynamic problem in three ways. First, price expectations are based on the information available at $h - z$ rather than the latest pre-dispatch price. Second, the battery observes s_{h-z} rather than the realized state of charge s_h in the target interval. Third, bids cannot respond to shocks realized after they have been committed.

Let $G_h^z(p_h | p_{h-z-1})$ denote the conditional distribution of p_h given the most recent observed price p_{h-z-1} . It is obtained by iterating the one-period transition kernels $F_h(p_h | p_{h-1})$:

$$G_h^z(p_h | p_{h-z-1}) = \int \cdots \int F_h(p_h | p_{h-1}) \prod_{j=1}^z dF_{h-j}(p_{h-j} | p_{h-j-1}), \quad G_h^0(p_h | p_{h-1}) = F_h(p_h | p_{h-1}).$$

State of charge must also be projected forward from s_{h-z} . Over z periods, each interval yields discharge, charge, or no dispatch, generating up to 3^z possible paths. These depend on the price process and interim committed bids:

$$\mathcal{B}_{h-z} = \{b_{h-z}^g, b_{h-z}^l, b_{h-z+1}^g, b_{h-z+1}^l, \dots, b_{h-1}^g, b_{h-1}^l\}.$$

The induced distribution of s_h is

$$P_h^z(s_h | s_{h-z}, p_{h-z-1}, \mathcal{B}_{h-z}) = \sum_{\{s_\tau\}_{\tau=h-z+1}^{h-1}} \prod_{j=-z}^{-1} P_{h+j}(s_{h+j+1} | s_{h+j}, b_{h+j}^g, b_{h+j}^l; p_{h-z-1}). \quad (9)$$

Compared to equation (4), expected profit under lead time requires z -period price and state-of-charge projections:

$$\begin{aligned} & \mathbb{E}_h[\pi(b_h^g, b_h^l, s_{h-z}, p_{h-z-1}, \mathcal{B}_{h-z})] \\ &= \sum_{s_h} \left\{ \int_{p_h \geq b_h^g} q_h^g(s_h) \times p_h dG_h^z(p_h) - \int_{p_h \leq b_h^l} q_h^l(s_h) \times p_h dG_h^z(p_h) \right. \\ & \quad \left. - \zeta \times [q_h^g(s_h)(1 - G_h^z(b_h^g)) + q_h^l(s_h)G_h^z(b_h^l)] \right\} \times P_h^z(s_h | s_{h-z}, p_{h-z-1}, \mathcal{B}_{h-z}). \end{aligned} \quad (10)$$

In period $h - z$, the battery chooses bids for interval h conditional on current state-of-charge s_{h-z} , previous-period price p_{h-z-1} , and interim bids $\mathcal{B}_{h-z}$. The value function is

$$\begin{aligned} & V_{h-z}(s_{h-z}, p_{h-z-1}, \mathcal{B}_{h-z}) \\ &= \max_{b_h^l \leq b_h^g} \{ \mathbb{E}_h[\pi(b_h^g, b_h^l, s_{h-z}, p_{h-z-1}, \mathcal{B}_{h-z})] \\ & \quad + \beta \mathbb{E}_h[V_{h-z+1}(s_{h-z+1}, p_{h-z}, \mathcal{B}_{h-z+1}) | b_h^g, b_h^l, s_{h-z}, p_{h-z-1}, \mathcal{B}_{h-z}] \}. \end{aligned} \quad (11)$$

4.3 Market Equilibrium

Using the optimal bids characterized in Sections 4.1 and 4.2, we embed battery bidding in

market equilibrium. Let non-battery electricity demand be $D_t(p_t)$ and renewable generation be R_t . Non-battery net demand is $X_t^{NB}(p_t) = D_t(p_t) - R_t$. Let $X_t^B(p_t)$ denote battery net charging load, which is positive when batteries charge and negative when they discharge.

Non-renewable generators supply net demand. We use $S_t(p_t)$ to represent their aggregate offer curve. The equilibrium price p_t^* clears the market:

$$X_t^{NB}(p_t^*) + X_t^B(p_t^*) = S_t(p_t^*). \quad (12)$$

Therefore, the last-minute-bidding equilibrium is defined as follows.

1. Battery i solves its dynamic bidding problem and submits bids according to equations (7) and (8), given the price process in equation (3).
2. Aggregate battery net charging load and non-battery net demand are supplied by non-renewable generation. This determines the market price according to equation (12).
3. Batteries have rational expectations over the equilibrium price process, so equation (3) holds true in equilibrium.

The equilibrium under bid lead time is defined analogously, except that batteries solve the lead-time dynamic problem. Our definition parallels a moment-based Markov Equilibrium (Ifrach and Weintraub, 2017), where each battery tracks the equilibrium electricity price process as the key aggregate moment.¹³

4.4 Model Discussion

Our model captures battery bidding incentives and equilibrium price responses. However, for tractability, we adopt several simplifying assumptions. We discuss their implications and justification.

Simplified bidding space

We assume that batteries submit a single generation bid price and a single load bid price for each 15-minute period, although in practice they can submit multi-unit bid functions with more than ten segments.¹⁴ To characterize typical bidding behavior, we analyze 2024 battery bids. We define a bid segment as *valid* if its price lies between -\$100 and \$1,000/MWh, as

¹³Several recent empirical papers use similar equilibrium definitions, such as Barwick et al. (2025), Garcia-Osipenko et al. (2025), Chen (2025), among others.

¹⁴In the data, however, no battery submitted a 15-minute bid function with more than 12 segments in 2024.

the probability of prices falling outside this range is approximately 0.1% in either tail. In addition, we merge adjacent segments whose bid quantities differ by less than 0.1 MW or whose bid prices differ by less than \$0.1/MWh. Appendix Figure A7 plots the distribution of valid bid segments across batteries and intervals. We find that 84% of generation bids contain no more than three valid segments. Tracking one generation bid price and one load bid price keeps the lead-time problem feasible by avoiding a much higher-dimensional state space.

We also assume that bid quantities q_i^g and q_i^l are exogenous and constant within each interval, although the data exhibit variation in quantities across periods. We decompose variation in bid quantities in Appendix Table A2. Battery fixed effects alone explain more than 80% of variation in bid quantities across batteries and 15-minute intervals. Adding battery-by-day or battery-by-time-of-day fixed effects further improves model fit, but only modestly. In contrast, the time variation alone, captured by day-by-interval fixed effects, explains only about 2% of the total variation. These results suggest that bid-quantity variation is driven primarily by cross-battery differences. Accordingly, time-invariant quantities abstract from smaller temporal variation, while simplifying the state space and facilitating the analysis of bid lead times.

Abstracting from ancillary service markets

Batteries in ERCOT also participate actively in day-ahead ancillary service markets. Because our primary focus, the bid lead time, is a feature specific to the real-time market, we abstract from batteries' participation in ancillary services in the model. Accordingly, we restrict the estimation sample to batteries that exclusively participate in the real-time market. Moreover, Appendix Figure A8 shows that as ancillary service markets become saturated, batteries increasingly rely on revenues from real-time energy arbitrage, with the share rising from less than 10% in 2020 to over 50% in 2025. Given our focus on the real-time market, our simulations are most relevant for a near-future grid in which ancillary service markets are saturated and batteries derive the majority of their revenues from real-time arbitrage, consistent with Butters et al. (2025). Recent work on ancillary services by Reynolds (2025) complements our analysis of battery operations and bidding behavior.

Ramping of fossil-fueled generators

We endogenize real-time electricity prices by allowing battery net discharge to shift net demand along the electricity supply curve. However, we assume that the electricity supply curve does not respond to changes in battery bidding behavior. This assumption reflects

our focus on short-run equilibria, where we compare alternative market designs for bidding while holding the composition of generation technologies fixed. As shown in Appendix Figure A9, coal and gas plants still account for a large share of ERCOT generation, while installed battery capacity remains small relative to system size. Consequently, changes in bidding behavior within the existing battery fleet are unlikely to meaningfully affect ramping decisions or induce generation mix changes.

Additional price predictors

Batteries may, in principle, exploit a rich set of information to form price forecasts and guide bidding decisions. Our model instead uses a parsimonious state space in which batteries forecast future prices using the most recent real-time price realization and combine this information with current state of charge to determine optimal bids. Importantly, as the lead-time problem above shows, the introduction of bid lead time rapidly expands the effective state space even when starting from this minimal set of state variables under last-minute bidding.

Our primary counterfactual comparison is between last-minute bidding and alternative regimes with positive bid lead times. While batteries also observe additional predictors such as day-ahead prices and renewable generation forecasts, these variables remain fully observable at the time bids are submitted, even under bid lead time. In this simplified information structure, the most recent real-time price captures the dynamically updating component of the battery’s information set. To assess its empirical validity, we compare forecast errors from equation (3) in the data and in model simulations (see Appendix Figure A10). Overall, the model closely replicates the distribution of forecast errors observed in the data.

5 Model Estimation and Simulation

Because most batteries operate in both the real-time energy market and the day-ahead ancillary service market, we first identify batteries that participate exclusively in the real-time energy market. Appendix Figure A14 summarizes this exercise. Among 269 batteries operating in ERCOT between March 2024 and November 2025, we identify 39 batteries that participate meaningfully and almost exclusively in the real-time market for at least one entire season.¹⁵ We focus on this subset for estimation.

¹⁵We partition the sample period (March 2024–November 2025) into seven three-month seasons: Spring 2024 (Mar–May), Summer 2024 (Jun–Aug), Fall 2024 (Sep–Nov), Winter 2024–25 (Dec–Feb), Spring 2025, Summer 2025, and Fall 2025. Each battery is assigned a single season for estimation; when multiple qualify,

5.1 Empirical price process

We estimate the electricity price process from real-time prices at each battery’s grid location (its settlement point, the node at which its dispatch is priced) during its assigned three-month season, using a semiparametric specification that preserves the empirical distribution of shocks and their within-day dependence.

$$p_{d,h} = \phi_h^s p_{d,h-1} + \xi_h^s + \epsilon_{d,h}$$

We use $\epsilon_{d,h}$ to denote the residual at period h from historical day d . The superscript s indexes the grid location and season pair; we estimate ϕ_h^s and ξ_h^s via OLS separately for each such pair and each 15-minute period, then construct daily residual blocks such that for each historical day d , residuals are collected into $\epsilon_d = (\epsilon_{d,1}, \dots, \epsilon_{d,96})$. We resample the residual vectors $\epsilon_d = (\epsilon_{d,1}, \dots, \epsilon_{d,96})$ with replacement in the remaining estimation steps to preserve within-day correlation and spike clustering. This semiparametric approach preserves the intraday price distribution, which parametric error distributions often fail to capture. Overall, we fit the key intraday statistics of prices well, as shown in Appendix Figure A13.

5.2 Cycling cost and round-trip efficiency of batteries

We calibrate each battery’s round-trip efficiency δ directly from the data by imposing zero net change in state of charge over the three-month sample period. We set the annual discount factor to 0.85, which implies a per-15-minute discount factor of $\beta = 0.85^{\frac{1}{365 \times 96}}$. The unit cycling cost ζ is the key parameter of interest in the estimation. Together with the expected price process, it determines the optimal cycling behavior of each battery, in particular the number of charge–discharge cycles per day. Holding cycling costs fixed, greater price volatility and a wider support of expected prices increase arbitrage opportunities, leading batteries to cycle more frequently. Conversely, for a given price process, a higher unit cycling cost reduces cycling frequency by making charge–discharge activity more costly in terms of battery degradation. Because the expected price process is estimated in the first stage, the observed number of cycles identifies the cycling cost for each battery.

We estimate the unit cycling cost ζ using the simulated method of moments (SMM), matching model-predicted battery operations to observed behavior. Simulating these operations requires solving the dynamic programming problems in equations (6) and (11). Because the AR(1) coefficients ϕ_h^s and ξ_h^s vary across the 96 within-day periods but repeat each day within a season, the dynamic problem is non-stationary within a day yet stationary across

we prioritize the most recent summer, then spring, fall, and winter.

days. We exploit this structure with a two-tier solver. The inner loop performs backward induction across the 96 fifteen-minute periods of a representative day: at each lagged price and state-of-charge cell, we evaluate the closed-form bid prices implied by the first-order conditions, compute dispatch probabilities and revenues from the empirical one-step price transition matrix, and update the value function. The outer loop iterates this within-day map, which links the start-of-day value to the end-of-previous-day value across day boundaries until the bid policies converge in sup norm.

To make this iteration computationally tractable, we discretize the state space. For each 15-minute period h , we discretize the lagged price into $N_p = 30$ nodes placed at quantiles of the empirical price distribution at that period, giving concentrated resolution at the tails where bidding decisions are most sensitive. State of charge is discretized uniformly into $4D + 1$ grid points $\{0, \frac{1}{4D}, \frac{2}{4D}, \dots, 1\}$, where D is the battery’s duration in hours; the grid step matches the fraction of capacity charged or discharged in one 15-minute interval at the battery’s nameplate power. Within the inner loop, we evaluate the optimal bid prices and continuation value at every combination of the N_p price nodes and $4D + 1$ SoC nodes.

We use r to index simulation draws. For a given ζ , the model generates simulated quantities $q_{dhr}(\zeta)$ (e.g., energy charged or discharged) that depend on the optimal bidding policies implied by the dynamic model. Let $\bar{q}_h$ denote the empirical average quantity in interval h . For each simulated day d and simulation draw r , we define the discrepancy between simulated and observed quantities across all intraday periods as

$$\ell_{dr}(\zeta) = [\sum_h (q_{dhr}(\zeta) - \bar{q}_h)]^2.$$

We simulate $N_d = 300$ days and $R = 1000$ independent price paths, and construct the sample objective function as the average loss:

$$Q(\zeta) = \frac{1}{N_d R} \sum_{d=1}^{N_d} \sum_{r=1}^R \ell_{dr}(\zeta) = \frac{1}{N_d R} \sum_{d=1}^{N_d} \sum_{r=1}^R [\sum_h (q_{dhr}(\zeta) - \bar{q}_h)]^2.$$

We estimate the cycling cost parameter as $\hat{\zeta} = \arg \min_{\zeta} Q(\zeta)$, using simulated battery paths generated from the estimated price process and the model-implied bidding policies.

5.3 Estimation results and model fit

The cycling cost estimates are shown in Table 1. These estimates are constrained by the estimated price process and the observed number of daily cycles in the data. Most batteries in our sample have cycling costs between \$3 and \$30/MWh, broadly consistent with engineering benchmarks such as He et al. (2020). One notable exception is Russek

Street BESS, for which we estimate a substantially higher cycling cost. This reflects the combination of persistently high electricity prices at its settlement point and a relatively low observed cycling frequency, which together imply that the battery forgoes profitable arbitrage opportunities and is consistent with a high marginal cycling cost.

As shown in Figure 5, we find that cycling costs are higher for batteries with lower average daily discharge relative to capacity. Batteries with larger nameplate power capacity (MW) tend to exhibit slightly lower cycling costs, while we find little to no correlation between cycling costs and battery durations.

We assess model fit by simulating the operations of sampled batteries and comparing them with the observed data. We focus on one battery, Casl Gap, a one-hour battery with 10 MW of nameplate power capacity.¹⁶ We calibrate its round-trip efficiency to 0.90 so that total charging and discharging are balanced over the sample. Appendix Figure A15 compares simulated and observed charging and discharging patterns. Overall, the model captures the daily charging and discharging patterns well.

5.4 Model Simulations under Bid Lead Time

State-space representation

Solving the dynamic bidding problem with bid lead time is empirically challenging because the battery must keep track of all interim bid prices that have already been committed in $\mathcal{B}_{h-z}$. This substantially expands the state space. For example, with a 75-minute bid lead time, $\mathcal{B}_{h-z}$ contains committed charge and discharge bids for five intervening 15-minute intervals, adding 10 state variables.¹⁷ To keep the dynamic problem tractable, we replace the full state vector with a lower-dimensional representation, $\Phi(s_{h-z}, p_{h-z-1}, \mathcal{B}_{h-z})$, which summarizes the payoff-relevant information needed to evaluate the profit function in equation (10). The key idea is that, rather than tracking s_{h-z} and the entire vector of committed bids $\mathcal{B}_{h-z}$ separately, we track the conditional distribution of the state of charge s_h at the time the bid becomes active.

Because batteries are assumed to charge and discharge at their nameplate power capacity, a battery with duration (D) hours can move across $(4D)$ discrete 15-minute charge

¹⁶Official records classify Casl Gap as a 4.2-hour battery. However, its observed charging and discharging patterns are closer to one-hour operation.

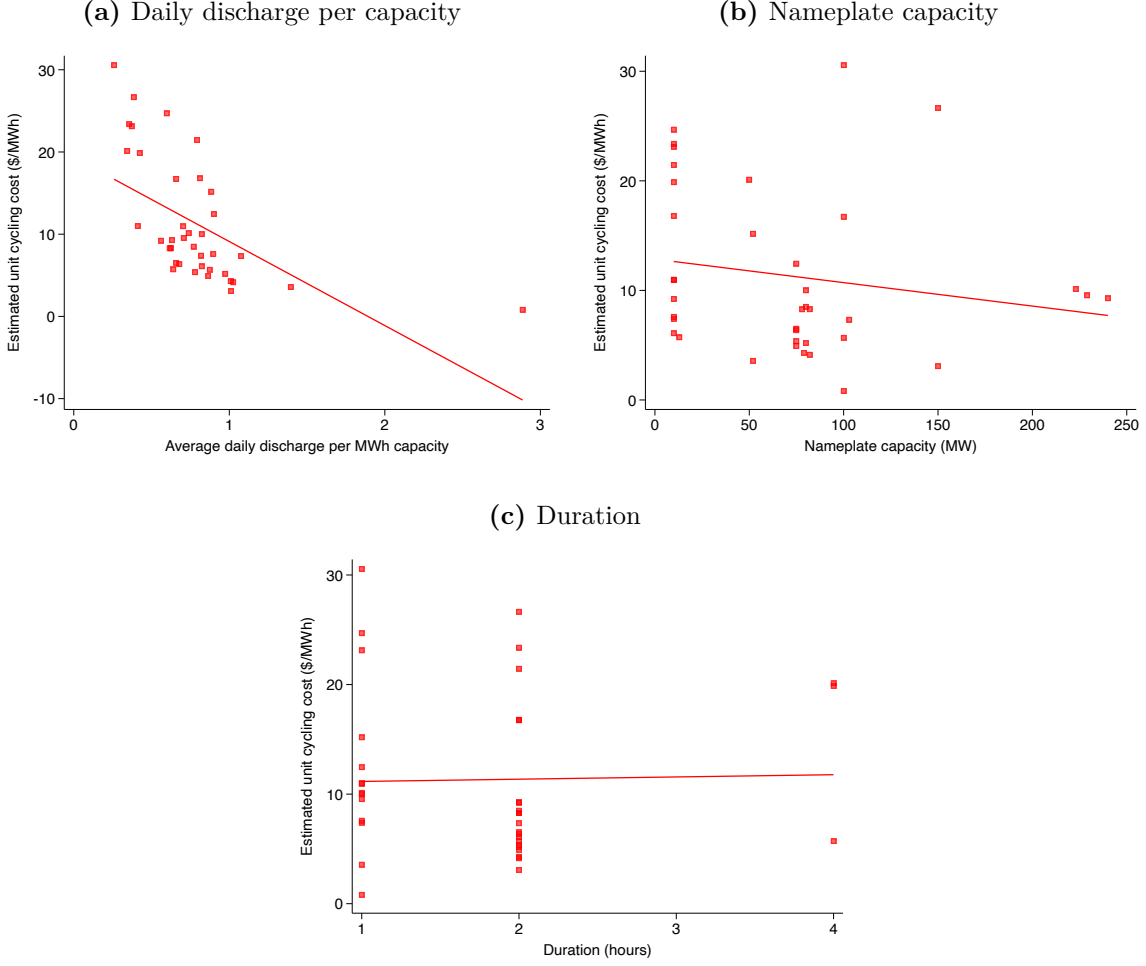
¹⁷ If each state variable is discretized using five grid points, the full state space would contain 5^{12} , or approximately 244 million, combined grid points.

Table 1: Estimated Cycling Costs for ERCOT Battery Storage Units

Resource name	Capacity (MW)	Duration (h)	Cycling cost (\$/MWh)	SE
Andromeda Solar 1	82	2	8.3	0.01
Andromeda Solar 2	78	2	8.3	0.01
Angelo Storage	100	2	16.7	0.03
Anole	240	2	9.3	0.01
Bexar ESS	100	1	30.6	0.03
Big Star	80	1	10.0	0.03
Bright Arrow 1	52	1	15.2	0.01
Bright Arrow 2	52	1	3.6	0.01
Burksol	100	2	5.7	0.01
Callisto I	223	1	10.1	0.01
Castle Gap	10	4	19.9	0.03
Corazon (Guajillo)	229	1	9.5	0.01
Danish Fields 1	75	2	6.4	0.01
Danish Fields 2	75	2	6.5	0.01
Desert Willow	150	2	3.1	0.01
Dogfish	75	1	12.5	0.03
Empire Central	10	1	23.1	0.02
Falfurrias	10	1	11.0	0.02
Fort Duncan	103	2	7.3	0.03
Fort Mason	10	1	24.7	0.02
Inertia	13	4	5.7	0.01
Jade Solar 1	79	2	4.3	0.00
Jade Solar 2	82	2	4.1	0.01
Jaguar Solar	150	2	26.7	0.04
Muenster	10	1	7.6	0.01
Myrtle 1	75	2	4.9	0.01
Myrtle 2	75	2	5.4	0.01
Padua 1	50	4	20.1	0.03
Palacios	10	1	11.0	0.02
Pirate	10	1	7.4	0.02
Prairie Creek	10	2	23.4	0.04
Radian Solar 1	80	2	8.5	0.01
Radian Solar 2	80	2	5.2	0.01
Rio Grande 2	10	2	9.2	0.01
Russek Street	10	1	95.1	0.32
Saragosa	10	2	21.5	0.05
Shamrock	100	1	0.8	0.00
Telview	10	2	6.1	0.01
Westover	10	2	16.8	0.02

Notes: This table shows the estimated unit cycling cost of 39 battery–season pairs that participate meaningfully and almost exclusively in the real-time market.

Figure 5: Cycling costs across batteries



Notes: This figure shows the estimated unit cycling costs of 39 battery–season pairs that participate meaningfully and almost exclusively in the real-time market, along with their correlations with average daily discharge per energy capacity (in MWh) in Panel (a), nameplate capacity (in MW) in Panel (b), and battery duration (in hours) in Panel (c). We drop Russek Street BESS since its estimated unit cycling cost is an outlier.

increments, corresponding to $(4D + 1)$ grid points including the empty and full states: $\mathcal{S} = (s_h^1, s_h^2, \dots, s_h^N) = \{0, \frac{1}{4D}, \frac{2}{4D}, \dots, 1\}$. We track its conditional distribution at the time the bid is submitted, and define

$$\boldsymbol{\tau}_h = (\tau_h^1, \tau_h^2, \dots, \tau_h^N), \quad \tau_h^n = \Pr(s_h = s^n).$$

Thus, $\boldsymbol{\tau}_h$ summarizes the battery’s belief about its state of charge when the realized s_h is not yet known because of the bid lead time.¹⁸ Accordingly, the relevant state can be written

¹⁸An alternative would be to track only the conditional mean, $\bar{s}_h = \mathbb{E}_{h-z}[s_h]$. However, this is generally insufficient because the profit function is nonlinear in s_{h-z} and $\mathcal{B}_{h-z}$. Tracking only the mean therefore discards information about the distribution of future states and misses the curvature of the profit function.

as

$$\Phi(s_{h-z}, p_{h-z-1}, \mathcal{B}_{h-z}) = \{\tau_h, p_{h-z-1}\}.$$

Dynamic bidding under lead time

To solve batteries' optimal bid prices under different lead times, we first construct the conditional distribution of p_h given the most recent price observed at submission p_{h-z-1} , $G_h^z(p_h | p_{h-z-1})$, by iterating forward the one-period price transition kernels $F_{h-k}(\cdot | \cdot)$ empirically. For each starting price node we draw 10^5 sample paths of $z+1$ -step-ahead prices from the estimated AR(1) model via the daily-block bootstrap of residuals (Section 5.1), which preserves within-day dependence across all intermediate periods, and use the simulated sample as the empirical G_h^z in the value-function recursion. To visualize price forecasts under bid lead time, we simulate future price distributions conditional on the mean realized price in each period. As shown in Appendix Figure A16, mean future prices under different lead times closely track mean future prices without lead time (Panel (a)), while the dispersion of projected prices increases substantially as lead time becomes longer (Panel (b)). Thus, bid lead time substantially increases the probability of tail price realizations in the projected price distribution.

The battery's belief about the conditional distribution of state of charge s_h is summarized by τ_h . The distribution τ_h evolves according to the battery's charging and discharging outcomes. Let T^g denote the transition matrix associated with discharge, T^l the transition matrix associated with charge, and I the identity matrix corresponding to no dispatch. Then the law of motion for beliefs can be written as

$$\tau_{h+1} = (1 - G_h^z(b_h^g))\tau_h T^g + G_h^z(b_h^l)\tau_h T^l + (G_h^z(b_h^g) - G_h^z(b_h^l))\tau_h I.$$

The value function can then be expressed as functions of the belief state τ_h .

$$V_{h-z}(\tau_h, p_{h-z-1}) = \max_{b_h^l \leq b_h^g} \mathbb{E}_h[\pi_h(b_h^g, b_h^l, \tau_h, p_{h-z-1})] + \beta \mathbb{E}_h[V_{h-z+1}(\tau_{h+1}, p_{h-z}) | b_h^g, b_h^l, \tau_h, p_{h-z-1}].$$

Similar to the last-minute bidding scenario, we solve the dynamic bidding under lead time with a two-tier solver. The main difference is that we now discretize the simplex of belief states τ_h , instead of state of charge s_h .

For this reason, we use the full distribution τ_h as the state variable.

State-of-charge-dependent bids

We next consider a policy allowing bids to depend on realized state of charge, a remedy discussed by the CAISO Market Surveillance Committee in Section 2. Instead of submitting one pair of bid prices b_h^g and b_h^l , batteries submit bid functions $b_h^g(s_h)$ and $b_h^l(s_h)$. Fully flexible functions $b_h^g(s_h)$ and $b_h^l(s_h)$ make the transition process for τ_h high-dimensional. To maintain tractability, we adopt a two-regime structure:

$$b_h^g(s_h) = \begin{cases} b_h^{g1}, & s_h < 0.5 \\ b_h^{g2}, & s_h \geq 0.5 \end{cases} \quad b_h^l(s_h) = \begin{cases} b_h^{l1}, & s_h \geq 0.5 \\ b_h^{l2}, & s_h < 0.5 \end{cases}$$

The low state-of-charge regime is $s_h < 0.5$, and the high state-of-charge regime is $s_h \geq 0.5$. Monotonicity requires $b_h^{g1} \geq b_h^{g2} \geq b_h^{l2} \geq b_h^{l1}$: low-SoC batteries become more conservative about discharging and high-SoC batteries about charging. The value function is

$$\begin{aligned} V_{h-z}(\tau_h, p_{h-z-1}) &= \max_{b_h^{g1}, b_h^{g2}, b_h^{l1}, b_h^{l2}} \mathbb{E}_h[\pi_h(b_h^{g1}(s_h), b_h^{g2}(s_h), b_h^{l1}(s_h), b_h^{l2}(s_h), \tau_h, p_{h-z-1})] \\ &\quad + \beta \mathbb{E}_h[V_{h-z+1}(\tau_{h+1}, p_{h-z}) | b_h^{g1}(s_h), b_h^{g2}(s_h), b_h^{l1}(s_h), b_h^{l2}(s_h), \tau_h, p_{h-z-1}] \end{aligned}$$

We quantify the welfare and bidding implications of this regime in Section 6.

Simulation of market equilibrium

With the price process and cycling cost estimated in stage one, we simulate the market equilibrium where battery policies and prices are jointly determined. Our approach operationalizes the equilibrium definition of Section 4.3: each battery bids optimally given its beliefs about the price process, the aggregate net dispatch of all batteries shifts the market-clearing condition along the non-renewable supply curve $S_t(p_t)$, and the resulting price process is consistent with those beliefs. We construct $S_t(p_t)$ directly from the ERCOT bid data, and solve for the fixed point via iteration over the price-process parameters.

We simulate the 39 batteries identified as operating meaningfully and almost exclusively in the real-time energy market. Each battery is characterized by its estimated round-trip efficiency $\hat{\delta}$, energy capacity k_i , duration D_i , and estimated cycling cost $\hat{\zeta}$.¹⁹ For the non-renewable supply curve $S_t(p_t)$, we construct period-specific supply curves $\{S_h(p)\}_{h=1}^{96}$ from

¹⁹We use system prices and aggregate battery output data from 2024 through December 4, 2025, the end of our sample as described in Section 2.3. We exclude days on which any 15-minute interval price exceeded \$1,000/MWh, since days with extreme prices are outliers that would otherwise distort the estimation of the price process parameters; this criterion removes 14 days from the sample.

ERCOT’s Security-Constrained Economic Dispatch (SCED) energy curves, averaging the price–quantity pairs at each 15-minute interval-of-day across all sample days. This per-interval averaging captures intraday variation in the marginal generator mix while smoothing idiosyncratic daily shocks.

Let p_t^m denote the observed market-clearing prices from the 2024–2025 data. We estimate the initial price-process parameters ξ_h and ϕ_h from the price data p_t^m ; these estimates serve as the starting values for the iterative procedure. Given the estimated price process, we solve the optimal bidding problem under last-minute bidding to obtain battery policies, and simulate battery net demand $X_t^{B,\text{sim}}$, defined as total battery load minus total battery generation at each interval t and taking empirical prices as given. We construct equilibrium net demand by inverting the supply curve at empirical prices: $X_t^{NB} = S_t(p_t^m) - X_t^{B,\text{sim}}$. The recovered X_t^{NB} series is held fixed across all counterfactual scenarios, so that differences in simulated prices between the last-minute bidding and 75-minute lead window cases reflect only the change in battery bidding behavior.²⁰

Let $\theta^{(0)} = \{(\xi_h^{(0)}, \phi_h^{(0)})\}_{h=1}^{96}$ denote the initial price-process parameter estimates from the observed price series. At iteration ℓ , let $\theta^{(\ell)} = \{(\xi_h^{(\ell)}, \phi_h^{(\ell)})\}_{h=1}^{96}$ denote the current price-process parameters. Holding X_t^{NB} fixed, we iterate over $\ell = 0, 1, 2, \dots$ as follows.

Step 1 (policy update): Given $\theta^{(\ell)}$, we solve each battery’s dynamic programming problem to obtain the optimal bidding policies for iteration ℓ , denoted by $b_{i,h}^{g,(\ell)}$ and $b_{i,h}^{l,(\ell)}$.

Step 2 (market clearing): Given the policies from Step 1, we simulate battery dispatch and market-clearing prices sequentially.²¹ At each interval t , the policies from iteration ℓ imply an aggregate battery net-demand schedule $X_t^{B,(\ell)}(p)$. At each interval, we find the next-iteration price $p_t^{(\ell+1)}$ satisfying the market-clearing condition (12), i.e.,

$$S_t(p_t^{(\ell+1)}) = X_t^{NB} + X_t^{B,(\ell)}(p_t^{(\ell+1)}).$$

The cleared price $p_t^{(\ell+1)}$ determines realized battery net demand $X_t^{B,(\ell)}(p_t^{(\ell+1)})$ and then serves as the lagged price on which batteries condition their bids in interval $t + 1$.

Step 3 (price-process update): We re-estimate the price-process parameters from

²⁰To reduce spurious spikes in X_t^{NB} arising from many batteries dispatching simultaneously, we apply Gaussian smoothing to $X_t^{B,\text{sim}}$ with bandwidth $\sigma = 2$ intervals over a five-interval window (± 30 minutes) centered on each dispatch interval before subtracting it from $S_t(p_t^m)$. We additionally clip X_t^{NB} to lie within the support of the supply curve. The supply-curve support in this implementation ranges from 2,873 MW to 55,549 MW.

²¹Batteries follow threshold-based dispatch rules. When batteries are simultaneously at the charge and discharge margins, charging bids are cleared first, and discharging bids are then cleared according to the standard dispatch rule.

the simulated equilibrium price series $\{p_t^{(\ell+1)}\}$. Denote the updated parameter vector by $\theta^{(\ell+1)} = \{(\xi_h^{(\ell+1)}, \phi_h^{(\ell+1)})\}_{h=1}^{96}$. We assess convergence using the mean absolute relative change across all elements k of θ , with a floor of 0.1 applied to the denominator to handle near-zero values:

$$\Delta^{(\ell+1)} = \frac{1}{K} \sum_{k=1}^K \frac{|\theta_k^{(\ell+1)} - \theta_k^{(\ell)}|}{\max(|\theta_k^{(\ell)}|, 0.1)}.$$

If $\Delta^{(\ell+1)} > 0.01$, we update via $\theta^{(\ell+1)} \leftarrow \theta^{(\ell)}$, set $\ell \leftarrow \ell + 1$, and return to Step 1. Otherwise, the fixed point has converged.

6 Counterfactual Results

We use the estimated structural model to answer three key questions. First, focusing on one representative battery, we examine how bid lead times shape bidding behavior and profitability, and how these effects interact with technology characteristics and price volatility. Second, we evaluate a policy alternative that allows batteries to submit state-of-charge-dependent bids, and assess its effectiveness in mitigating efficiency losses. Third, we embed these policy counterfactuals in a market equilibrium framework to quantify how bidding inefficiencies affect grid outcomes and, more broadly, the energy transition.

6.1 The impacts of bid lead time

We first investigate how lead time affects batteries' bidding strategies, net energy supply, and profits. We solve the generation and load bid prices under various bid lead time using parameters of Casl Gap, and then simulate operations of this one-hour batteries with 10 MW power capacity for 300 days by 1,000 times. The results are shown in Figure 6. Relative to last-minute bidding, batteries tend to submit lower generation bid prices throughout the day under longer lead time. This occurs because a larger dispersion in the projected price distribution increases the density of tails, which are the primary drivers of batteries' expected profits. Consequently, batteries are more likely to be discharged earlier in the day but supply less energy during peak-price periods, leading to a substantial deviation from optimal charging-discharging pattern under last-minute bidding.

Moreover, we document a sharp profit-lead horizon gradient as shown in Figure 7. The baseline simulated profit under zero lead time is \$10.4 per kW of installed capacity, which is close to the \$18.7 energy arbitrage revenue per kW of installed capacity as calculated by Modo Energy. Under 30-minute bid lead time as currently implemented in MISO, SPP, and ISO-NE, the average profit is reduced to \$8.6 per kW, which is only 82.6% of the baseline profit. Under 75-minute bid lead time as currently implemented in CAISO, the average

profit is reduced to \$7.9 per kW, which is only 75.7% of the baseline profit. These results highlight the importance of bidding horizon in shaping operational profits for batteries.

We further examine how the profit impacts of bid lead time vary with technology characteristics and price volatility. As shown in Figure 8, longer-duration batteries partially mitigate the distortions induced by bid lead time. The reduction in energy arbitrage profits declines from 24.3% for one-hour batteries to 11.7% for four-hour batteries, reflecting the greater inter-temporal flexibility of longer-duration storage. Given that more than 80% of installed capacity in CAISO consists of four-hour batteries, the adverse effects of bid lead time on profits are attenuated in practice, though they remain economically meaningful.

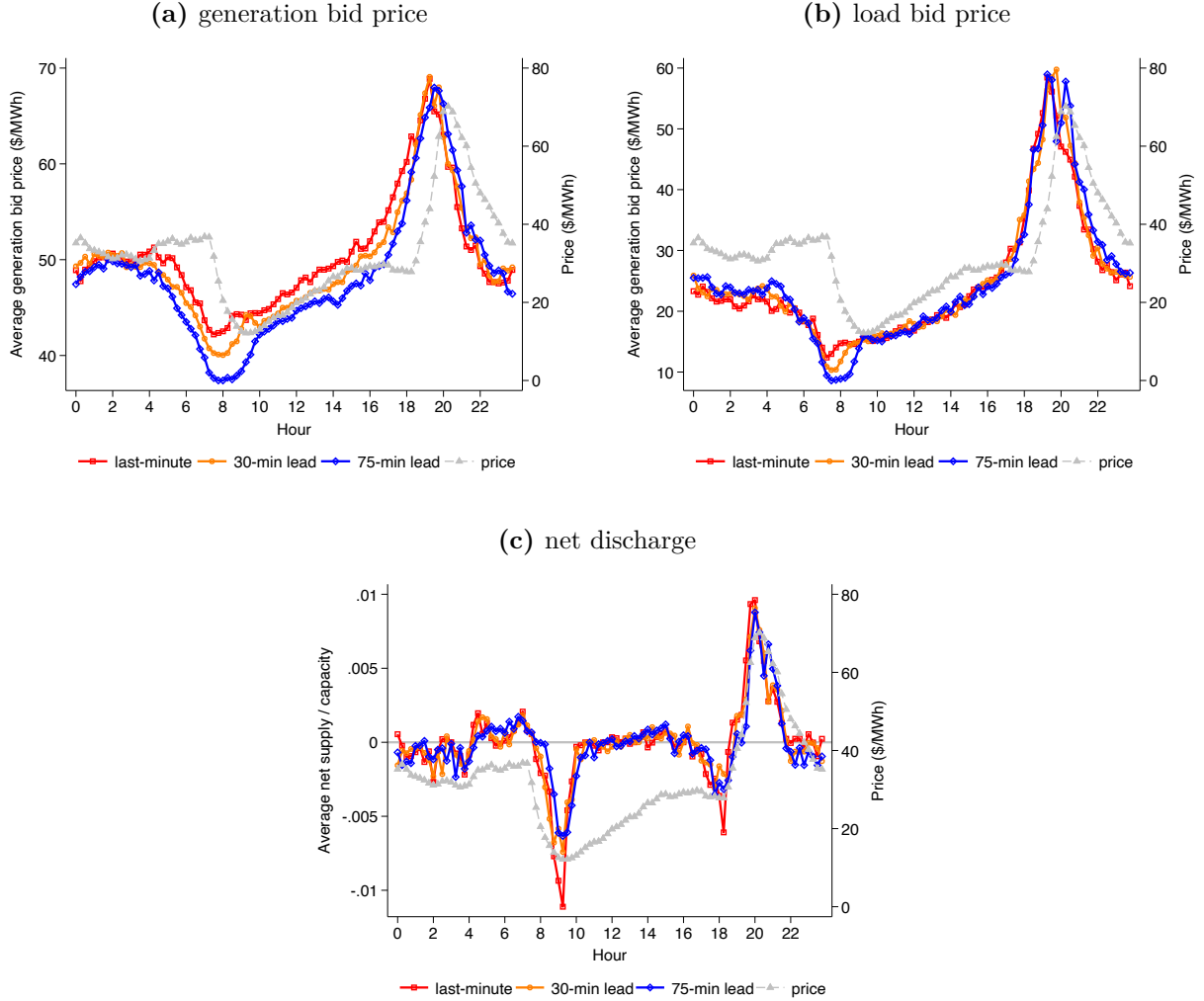
In contrast, the impact of bid lead time is amplified by greater price volatility. As shown in Panel (b), a 10% increase in price volatility raises profit losses from 24.3% to 28.8%. These results suggest that bid lead time becomes a more consequential market design friction as volatility increases, particularly in systems with higher renewable penetration where price fluctuations are more pronounced.

6.2 State-of-charge-dependent bids

We next evaluate the state-of-charge-dependent bidding regime introduced in Section 5.4, in which batteries submit bids $b_h^g(s_h)$ and $b_h^l(s_h)$ that depend on the realized state of charge. As a policy remedy discussed by the CAISO Market Surveillance Committee, this regime allows batteries to make contingent plans on the realized state of charge and mitigate the quantity uncertainty introduced by bid lead time.

We solve the generation and load bid prices using state-of-charge-dependent bids and under various bid lead time, and then simulate operations of the same identical one-hour batteries with 10 MW power capacity for 300 days by 1,000 times. The average generation bid prices, as shown in Panel (a) of Figure 9, differentiate substantially between the upper level (b_h^{g1}) and lower level (b_h^{g2}) with the regular generation bid prices closer to the lower level. This is intuitive since the potential opportunity cost of discharging will be much higher if the batteries have already been on low state of charge, and allowing state-of-charge dependent bids allow batteries to make contingent plans on realized state of charge and submit more sophisticated bids. Similarly, the average load bid prices are closer to the upper level (b_h^{l2}) with substantial level differences between the upper level (b_h^{l2}) and lower level (b_h^{l1}).

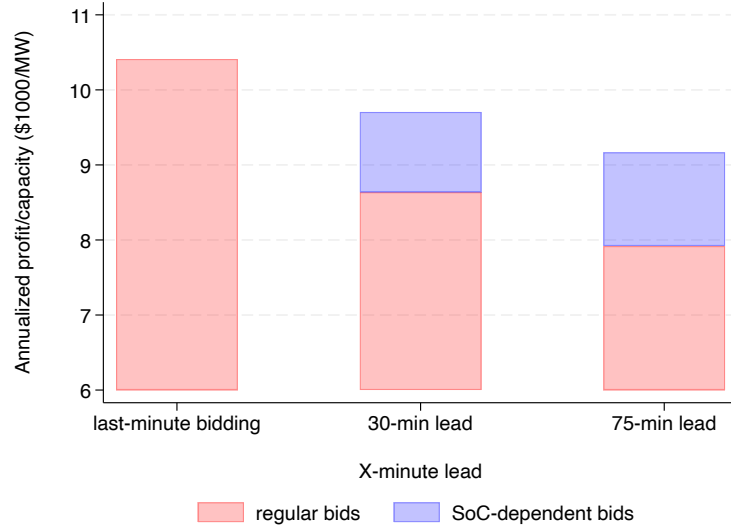
Figure 6: Bids, Energy Supply, Profits under Bid Lead Time



Notes: This figure plots the simulated generation bid prices, load bid prices, net energy supply, and profit at the 15-minute frequency for three scenarios: (1) last-minute bidding (red squares), (2) 30-minute bid lead time (orange circles), and (3) 75-minute bid lead time (blue diamonds). We further plot the real-time market prices in gray.

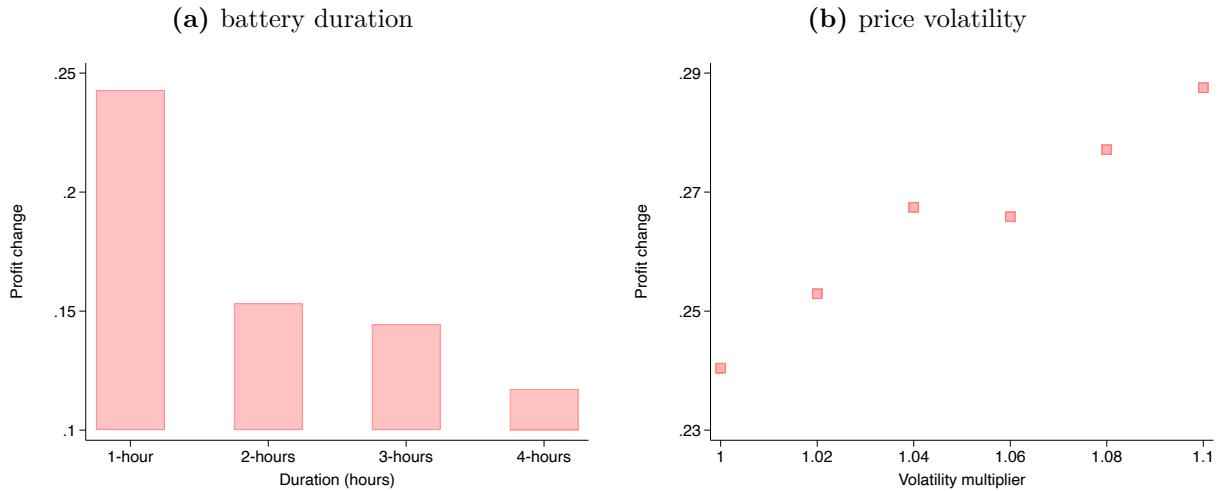
State-of-charge-dependent bids brings substantial improvement to batteries' operation profits, since they mitigate quantity uncertainty and allow contingent plans based on different realizations of state of charge. As shown in Figure 7, under 30-minute bid lead time, state-of-charge-dependent bids improve the average profit by 12.3%, and close the gap from the baseline profit by 10.2 percentage points (82.6% to 92.8%). Under 75-minute bid lead time, state-of-charge-dependent bids improve the average profit by 15.8%, and close the gap from the baseline profit by 12.0 percentage points (75.7% to 87.7%). The profit-lead horizon

Figure 7: Profits under Bid Lead Time and SoC-dependent Bids



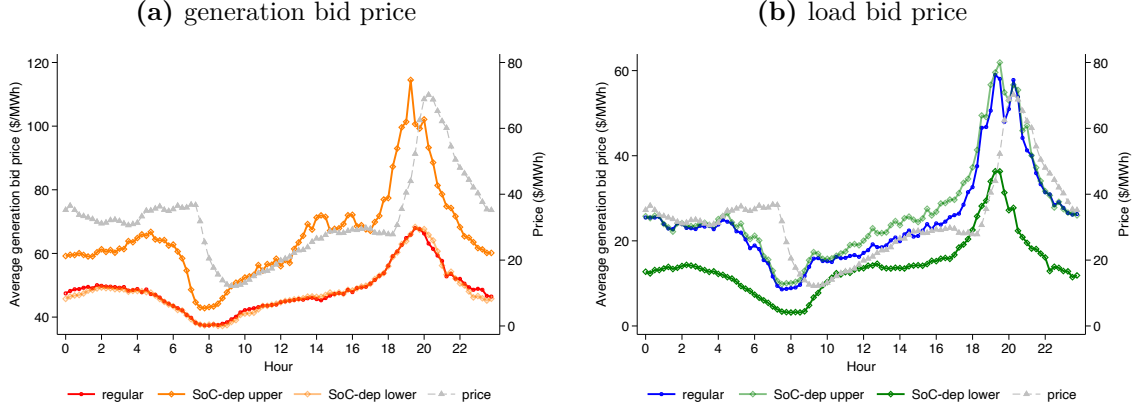
Notes: This figure plots the simulated profits under various bid lead time conditional on mean realized prices each period.

Figure 8: Profit Changes under Bid Lead Time: the Role of Durations and Volatility



Notes: This figure plots the profit impacts of bid lead time under different battery durations (Panel (a)) and price volatility (Panel (b)). The horizontal axis of Panel (b) denotes the multipliers we apply to the residuals estimated from equation (3).

Figure 9: Bids Prices Using SoC-dependent Bids



Notes: This figure plots the simulated generation bid prices and load bid prices under 30-minute bid lead time with regular bidding and SoC-dependent bids.

gradient is much flatter under state-of-charge-dependent bids. Therefore, allowing contingent bidding plans significantly lowers the uncertainty towards quantities, and improves batteries' operation profits.

6.3 Equilibrium consequences

We simulate the two bidding regimes, last-minute bidding and 75-minute bid lead time, in the market equilibrium framework of Section 4.3 to assess how bid lead time shapes electricity prices and aggregate welfare. We also report a no-battery counterfactual, in which battery net demand is held at zero and non-renewable supply alone clears the market, as a baseline for the overall effect of battery participation.

Price effects

Figure 10 visualizes the hour-of-day pattern of equilibrium prices and the price effects of removing bid lead time. The red line (left axis) plots the average market-clearing price by hour of day under no bid-lock, exhibiting the familiar ERCOT duck curve: prices average roughly \$25-\$30/MWh overnight and at midday, reach a modest morning shoulder around 7 a.m. (about \$40/MWh), and peak sharply around 8 p.m. at about \$56/MWh as solar generation tails off into the evening demand ramp. The blue bars (right axis) show the hour-by-hour difference between the 75-minute bid-lock and no bid-lock scenarios, with positive bars indicating that bid-lock raises prices relative to last-minute bidding and negative bars indicating the opposite.

Two patterns emerge. First, bid lead time systematically raises prices during the morning ramp (hours 6-7) and the evening peak (hours 17-21), by roughly \$1-\$2/MWh, with the largest gap of about \$1.7/MWh at hour 17. These are precisely the hours when batteries should be discharging most aggressively to shave peak demand, but the 75-minute bid commitment prevents them from responding to the realized price spike. Second, in overnight and late-evening hours (hours 1-5 and 22-24), prices under zero lead time are slightly higher (by \$0.5-\$2/MWh): under last-minute bidding, batteries charge more aggressively at low-price times, lifting the price floor. The net effect of removing bid lead time is therefore a compression of the daily price range, with peak prices falling and trough prices rising as equilibrium prices smooth toward the daily mean.

Table 2 reports the corresponding distribution of equilibrium prices across the three scenarios. The mean equilibrium price falls modestly under no bid-lock while the 99th percentile drops by about \$14.6/MWh, reflecting the same peak-shaving mechanism: bid lead time matters most for tail events when arbitrage is most valuable.

Welfare decomposition

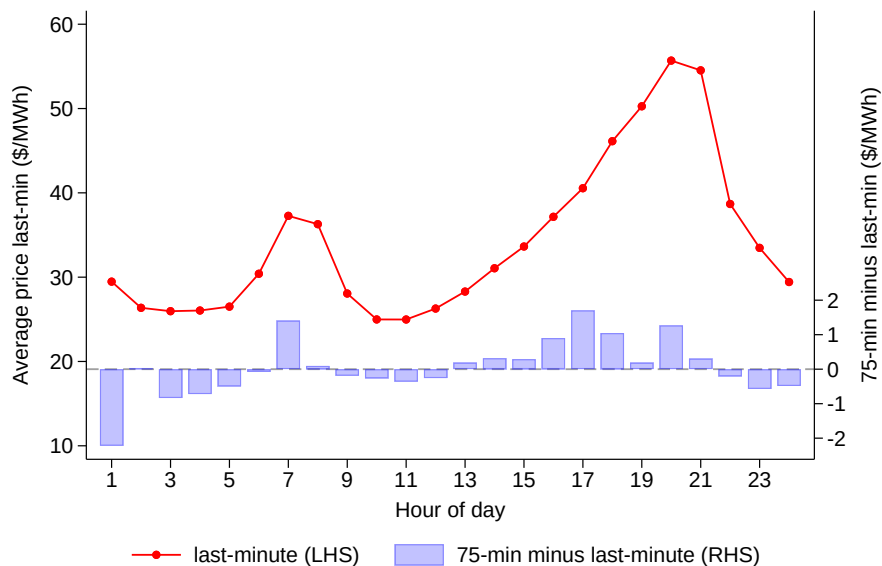
Table 3 reports average daily welfare components across the three scenarios. Removing the bid-lock window raises battery profits by roughly 7.9%, from \$14.6/day per MW of installed capacity under the 75-minute window to \$15.8/day, consistent with the individual-level profit gains documented in Section 6.1. Consumer surplus, measured here as the reduction in consumers' total energy expenditure (price times non-battery net demand) relative to the no-battery counterfactual, rises by roughly \$110,000 per day, or about 1.3%, when the bid-lock window is removed.

The lower prices under no bid-lock compress profits for other market participants. Average daily fossil profit falls modestly from \$33.7 million under the 75-minute window to \$33.6 million under last-minute bidding, a reduction of about \$110,000/day, with per-MWh margins essentially unchanged at roughly \$28.0. Wind revenue, by contrast, is essentially unchanged across the two regimes: we measure wind and solar revenues per MW of installed capacity as the capacity factor times the simulated market-clearing price, accumulated over each 15-minute interval, where capacity factors are derived from atmospheric conditions in the ERA5 reanalysis dataset. Daily wind revenue per MW of installed capacity is \$343.9 under the 75-minute window and \$344.6 under no bid-lock, a difference of less than \$1/MW/day.

Solar revenue is similarly close to flat across the two regimes (\$184.3 under no bid-lock versus \$186.9 under the 75-minute window), if anything slightly more favorable under the 75-minute window. The temporal arbitrage that batteries perform within a day — charging

during midday hours when prices are low and discharging during peak evening hours — tends to be particularly complementary with solar generation, which is high during the day and offline in the evening. Under no bid-lock, batteries are more aggressive about cycling as they are more certain that they can time their charge-discharge cycles precisely around price trends. This more aggressive midday arbitrage modestly compresses solar revenue by sharpening competition with solar during the hours when solar is most productive, though the magnitude is small relative to the redistribution from fossil generators.

Figure 10: Average Price by Interval under Bidding Counterfactuals



Notes: This figure plots equilibrium prices under two bidding regimes by hour of day. The blue line (left axis) shows the average market-clearing price under no bid-lock (last-minute bidding). The red bars (right axis) show the hour-by-hour difference between the 75-minute bid-lock and no bid-lock scenarios (positive = 75-minute window clears at a higher price). Averages are computed over all days in the sample.

Table 2: Summary Statistics of Simulated Market Prices

	No batteries	Last-minute bidding	75-min lead
5th percentile (\$/MWh)	4.8	4.9	4.8
Median (\$/MWh)	22.5	22.5	22.6
Mean (\$/MWh)	40.7	34.2	34.3
95th percentile (\$/MWh)	91.0	85.3	85.5
99th percentile (\$/MWh)	305.7	234.8	249.4

Notes: This table reports the 5th, 50th (median), 95th, and 99th percentiles together with the mean of simulated 15-minute real-time market prices under three scenarios: no batteries (battery net demand held at zero so non-renewable supply alone clears the market), last-minute bidding (zero bid-lock window), and a 75-minute bid-lock window (the bid-lock currently implemented in CAISO). All values are in \$/MWh.

Emissions and total surplus

We compute emissions by constructing an emissions supply curve that pairs each fossil plant’s position in the merit order with its CO₂ emission rate, yielding total CO₂ output (in lbs/MWh) as a function of aggregate non-renewable dispatch. At each simulated interval, we map the non-battery generation level onto this curve to recover total emissions, then convert to short tons. The difference in daily emissions between the two bidding regimes is negligible: daily emissions are 308.8 kiloton under both the 75-minute window and no bid-lock, essentially identical at the precision reported in the table. Applying a social cost of carbon of \$185 per short ton, the emissions channel contributes a negligible amount on the order of a few thousand dollars per day at most to the welfare comparison between regimes. Summing consumer surplus, battery profits, and fossil profits and netting out the social cost of emissions, the change in daily total surplus relative to the no-battery counterfactual is \$49,760 under the 75-minute window and \$56,540 under no bid-lock, which brings a net welfare gain of roughly \$6,800 per day from removing the bid-lock window. The gain reflects higher consumer surplus and higher battery profits, partially offset by lower fossil profits and essentially unchanged emissions. Taken together, these results indicate that the primary market-level consequence of removing bid lead time is a transfer from fossil producers to batteries and consumers, amounting to roughly \$110,000/day in consumer surplus alone.

Table 3: Average Daily Market Outcomes and Surplus Effects

	No batteries	Last-minute bidding	75-min lead
<i>Panel A: Market Outcomes</i>			
Fossil generation (GWh/day)	1,024.50	1,024.70	1,024.73
Fossil generation cost (M\$/day)	7.14	7.04	7.05
Emissions (kton/day)	308.65	308.78	308.78
Social cost of emissions (M\$/day)	57.10	57.12	57.12
<i>Panel B: Surplus Effects</i>			
Consumer surplus change (M\$/day)	0.00	8.85	8.74
Battery profit (\$/MW/day)	0.00	15.76	14.61
Fossil profit (M\$/day)	42.37	33.58	33.69
Fossil profit margin (\$/MWh)	34.48	28.03	28.07
Wind profit (\$/MW/day)	423.56	344.55	343.89
Solar profit (\$/MW/day)	204.95	184.33	186.94
Total surplus change (k\$/day)	0.00	56.54	49.76

Notes: This table shows the key statistics of the market under three scenarios: no bid-lock window, 75-minute bid-lock window, and no batteries.

7 Conclusion

Despite growing economic interests in energy storage, limited attention has been paid to how batteries operate in electricity markets, how efficiently they bid as energy arbitrageurs, and how bidding (in)efficiencies affect market equilibrium and the broader energy transition. This paper investigates a key market design feature of real-time electricity auctions that is particularly relevant to battery bidding: the bid lead time, which is a common design in the wholesale electricity market around the world and varies across markets. We compile a comprehensive data set covering all utility-scale batteries in ERCOT from 2018 to 2025, and we access the real-time bidding and settlement information for every 15 minutes. Using these high-frequency data, we document three key data patterns. First, we find that batteries are actively bidding into the real-time energy market, charging during low-price periods and discharging during high-price periods to maximize profits. Batteries submit high generation bid prices to avoid being dispatched too early and to preserve the high option value of stored energy. Second, we find that batteries respond to unexpected market shocks by quickly adjusting the generation bid prices. Third, since accurate prediction of future prices is a key prerequisite for batteries to earn profits, we demonstrate a critical role of recent price

information in reducing forecast errors. Motivated by these empirical facts, we develop an equilibrium model of electricity markets, which embeds a dynamic model of battery bidding. We then introduce bid lead time as a market design constraint such that bids for a given interval must be submitted several periods in advance. Overall, we find that bid timing is a quantitatively important dimension of market design: even relatively short lead times generate meaningful inefficiencies in storage operations and distort equilibrium outcomes in electricity markets.

References

- Allcott, H. (2012). Real-time pricing and electricity market design. *NBER Working paper*.
- Andrés-Cerezo, D. and Fabra, N. (2023). Storing power: Market structure matters. *The RAND Journal of Economics*, 54(1):3–53.
- Andrés-Cerezo, D. and Fabra, N. (2025). Storage and renewable energies: Friends or foes. Technical report, Working Paper.
- Athey, S., Levin, J., and Seira, E. (2011). Comparing open and sealed bid auctions: Evidence from timber auctions. *The Quarterly Journal of Economics*, 126(1):207–257.
- Ausubel, L. M., Cramton, P., Pycia, M., Rostek, M., and Weretka, M. (2014). Demand reduction and inefficiency in multi-unit auctions. *The Review of Economic Studies*, 81(4):1366–1400.
- Back, K. and Zender, J. F. (1993). Auctions of divisible goods: On the rationale for the treasury experiment. *The Review of Financial Studies*, 6(4):733–764.
- Barwick, P. J., Kalouptsi, M., and Zahur, N. B. (2025). Industrial policy implementation: Empirical evidence from china’s shipbuilding industry. *Review of Economic Studies*, 92(6):3611–3648.
- Bergemann, D. and Morris, S. (2013). Robust predictions in games with incomplete information. *Econometrica*, 81(4):1251–1308.
- Borenstein, S., Bushnell, J. B., and Wolak, F. A. (2002). Measuring market inefficiencies in California’s restructured wholesale electricity market. *American Economic Review*, 92(5):1376–1405.
- Bushnell, J. B., Mansur, E. T., and Saravia, C. (2008). Vertical arrangements, market structure, and competition: An analysis of restructured U.S. electricity markets. *American Economic Review*, 98(1):237–266.
- Butters, R. A., Dorsey, J., and Gowrisankaran, G. (2025). Soaking up the sun: Battery investment, renewable energy, and market equilibrium. *Econometrica*, 93(3):891–927.
- Chaves, I. N. and Ichihashi, S. (2024). Auction timing and market thickness. *Games and Economic Behavior*, 143:161–178.

- Chen, L. (2025). The dynamic efficiency of policy uncertainty: Evidence from the wind industry.
- Cicala, S. (2015). When does regulation distort costs? lessons from fuel procurement in U.S. electricity generation. *American Economic Review*, 105(1):411–444.
- Cicala, S. (2022). Imperfect markets versus imperfect regulation in U.S. electricity generation. *American Economic Review*, 112(2):409–441.
- Fabra, N. and Reguant, M. (2014). Pass-through of emissions costs in electricity markets. *American Economic Review*, 104(9):2872–2899.
- Garcia-Osipenko, M., Vreugdenhil, N., and Zahur, N. B. (2025). Endogenous rigidities and capital misallocation: Evidence from containerships. *Available at SSRN 5217587*.
- GridLab (2026). The storage gap: Deployment challenges in the east. Technical report, GridLab.
- He, G., Kar, S., Mohammadi, J., Moutis, P., and Whitacre, J. F. (2020). Power system dispatch with marginal degradation cost of battery storage. *IEEE Transactions on Power Systems*, 36(4):3552–3562.
- Hendricks, K., Sorensen, A., and Wiseman, T. (2025). Efficiency in dynamic auctions with endogenous matching.
- Holland, S. P., Mansur, E. T., and Yates, A. J. (2024). Electrification in the long run. Technical report, Tech. rep. National Bureau of Economic Research.
- Hortaçsu, A. and Kastl, J. (2012). Valuing dealers’ informational advantage: A study of Canadian treasury auctions. *Econometrica*, 80(6):2511–2542.
- Hortaçsu, A., Luco, F., Puller, S. L., and Zhu, D. (2019). Does strategic ability affect efficiency? evidence from electricity markets. *American Economic Review*, 109(12):4302–4342.
- Hortaçsu, A. and McAdams, D. (2010). Mechanism choice and strategic bidding in divisible good auctions: An empirical analysis of the Turkish treasury auction market. *Journal of Political Economy*, 118(5):833–865.
- Hortaçsu, A. and Perrigne, I. (2021). Empirical perspectives on auctions. In *Handbook of Industrial Organization*, volume 5, pages 81–175. Elsevier.

- Hortaçsu, A. and Puller, S. L. (2008). Understanding strategic bidding in multi-unit auctions: a case study of the texas electricity spot market. *The RAND Journal of Economics*, 39(1):86–114.
- Ifrach, B. and Weintraub, G. Y. (2017). A framework for dynamic oligopoly in concentrated industries. *The Review of Economic Studies*, 84(3):1106–1150.
- Ito, K. and Reguant, M. (2016). Sequential markets, market power, and arbitrage. *American Economic Review*, 106(7):1921–1957.
- Jha, A. and Wolak, F. A. (2023). Can forward commodity markets improve spot market performance? evidence from wholesale electricity. *American Economic Journal: Economic Policy*, 15(2):292–330.
- Karaduman, Ö. (2021). Economics of grid-scale energy storage.
- Kastl, J. (2011). Discrete bids and empirical inference in divisible good auctions. *The Review of Economic Studies*, 78(3):974–1014.
- Kirkpatrick, A. J. (2020). Estimating congestion benefits of batteries for unobserved networks: A machine learning approach.
- Klemperer, P. (2004). *Auctions: Theory and Practice*. Princeton University Press.
- Lamp, S. and Samano, M. (2022). Large-scale battery storage, short-term market outcomes, and arbitrage. *Energy Economics*, 107:105786.
- Mansur, E. T. (2008). Measuring welfare in restructured electricity markets. *The Review of Economics and Statistics*, 90(2):369–386.
- Milgrom, P. (2004). *Putting Auction Theory to Work*. Cambridge University Press.
- Milgrom, P. R. and Weber, R. J. (1982). A theory of auctions and competitive bidding. *Econometrica*, 50(5):1089–1122.
- Palley, T., Palley, A., Butters, R. A., and Cetin, O. (2024). Charging into the future: Intermittency and the role of energy storage in the transition to a renewable electric grid. *Available at SSRN 4849063*.
- Reguant, M. (2014). Complementary bidding mechanisms and startup costs in electricity markets. *The Review of Economic Studies*, 81(4):1708–1742.

- Reynolds, S. S. (2025). *Keeping the Lights On: Battery Storage, Operating Reserves, and Electricity Scarcity Pricing*.
- Wilson, R. (1979). Auctions of shares. *The Quarterly Journal of Economics*, 93(4):675–689.
- Wolak, F. A. (2003). Identification and estimation of cost functions using observed bid data: An application to electricity markets. *Advances in Economics and Econometrics: Theory and Applications, Eighth World Congress*, 2:133–169.
- Xu, B., Zhao, J., Zheng, T., Litvinov, E., and Kirschen, D. S. (2017). Factoring the cycle aging cost of batteries participating in electricity markets. *IEEE Transactions on Power Systems*, 33(2):2248–2259.

Online Appendix for

Bid Timing and Market Efficiency in the Energy Transition

Hunt Allcott Luming Chen Julia Park

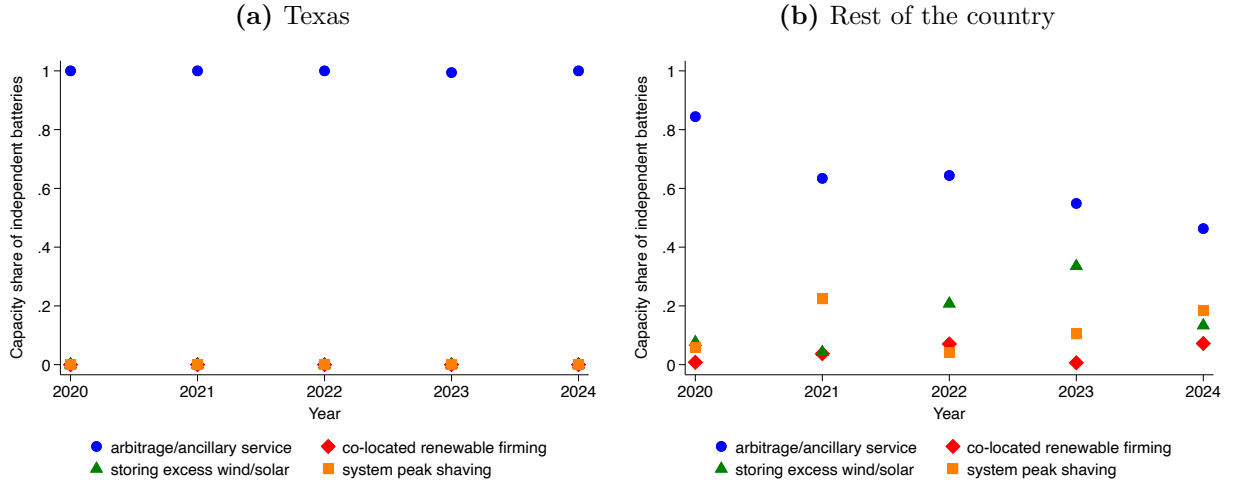
A Additional Figures and Tables

Table A1: Real-Time Bid Lead Times and Bid Commitment Lengths Across Wholesale Electricity Markets

Market	RT bid lead time	RT bid commitment length
<i>Panel A: US ISO/RTOs</i>		
ERCOT (Texas)	≈ 0 min [†]	5 min [‡]
CAISO (California)	75 min	60 min
MISO (Midcontinent)	30 min	60 min
SPP (Southwest Power Pool)	30 min	60 min
ISO-NE (New England)	30 min	60 min
PJM (Mid-Atlantic)	65 min	60 min
NYISO (New York)	75 min	60 min
<i>Panel B: Selected International Markets</i>		
AEMO (Australia, NEM)	5 min	5 min
GB (Elexon/BMRS)	60 min	30 min
Nord Pool (intraday)	5–60 min [§]	15–60 min [§]

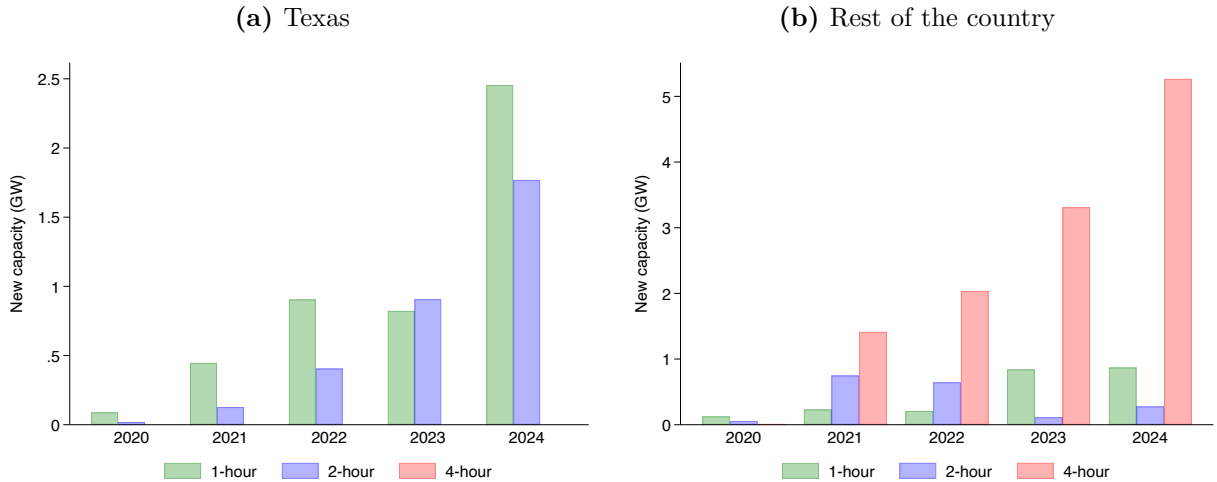
Notes: “RT bid lead time” is the gate-closure time before the start of the delivery period after which bids must remain fixed. “RT bid commitment length” is the duration over which a single bid applies once submitted. ERCOT allows energy offer curves to be updated during the operating period, effectively eliminating bid lead time; the pre-reform Adjustment Period implied a small but non-zero lead time. In some markets, gate closure is currently as short as 5 minutes before delivery in several zones. Cells reflect the rules most relevant for storage participating in the real-time energy market; details vary by resource type and may differ for ancillary services or under ongoing reforms. International entries should be verified against the latest market rulebooks.

Figure A1: Capacity Shares of Independent Batteries



Notes: This figure plots the capacity shares of batteries that are independent (i.e., not coupled with another generator) and that are not supporting other generation units from 2018 to 2023. A battery directly supports another unit (mostly solar) if it is intended for dedicated generator firming or storing excess generation of other units. We plot the capacity shares in Texas (Panel (a), the focus of this paper) and in the rest of the country (Panel (b)). The data source is EIA-860.

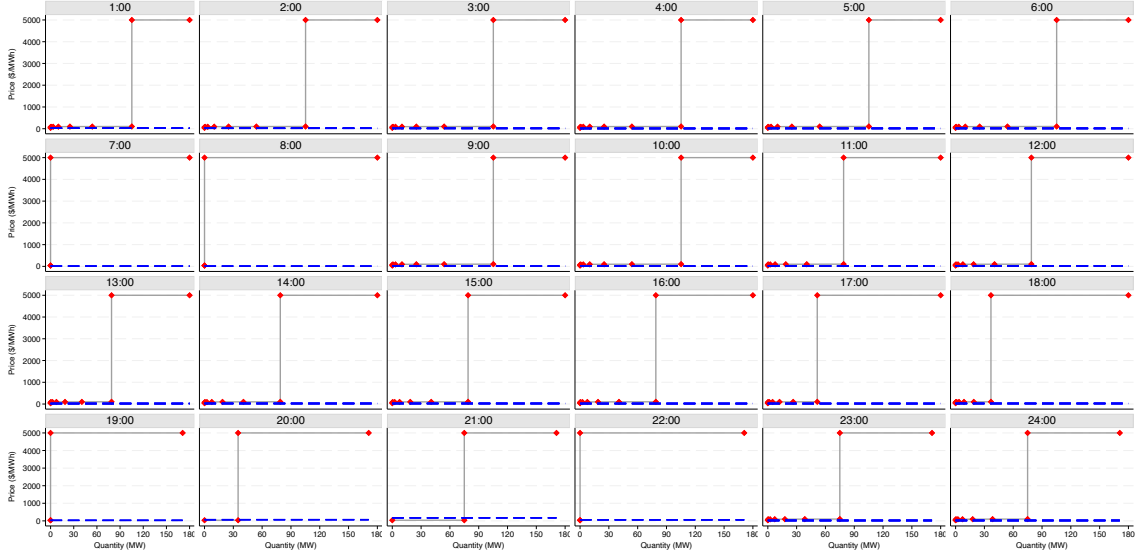
Figure A2: New Capacity of Batteries by Durations



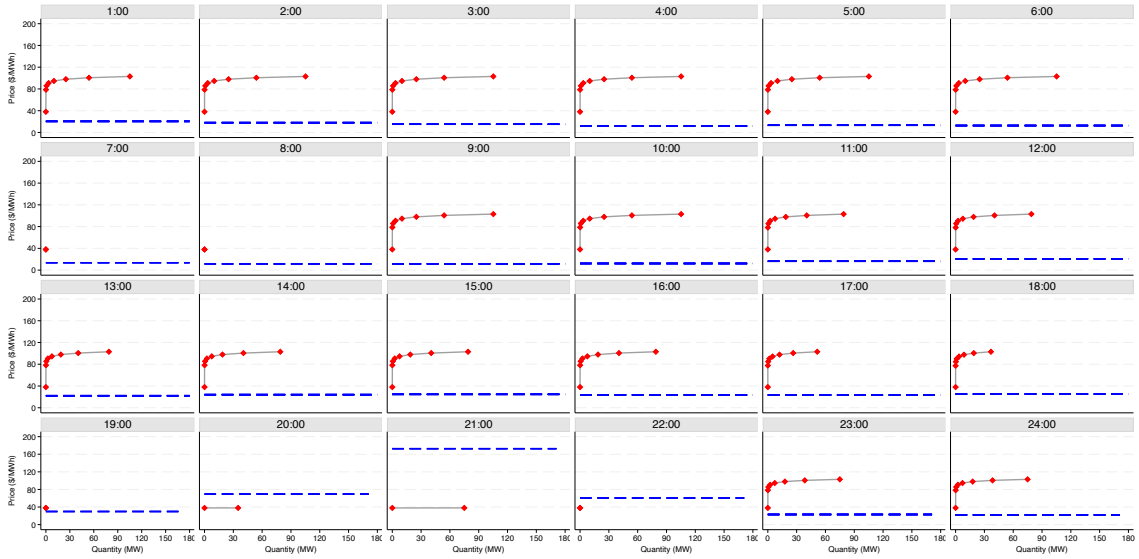
Notes: This figure plots the new capacity of batteries by durations (1-hour batteries, 2-hour batteries, 4-hour batteries) in Texas (Panel (a)) and in the rest of the country (Panel (b)) from 2018 to 2023. A 1-hour battery could maintain its full power output for 1 hour. The data source is EIA-860.

Figure A3: Generation Bids of Turquoise Storage on August 18, 2024

(a) raw pattern



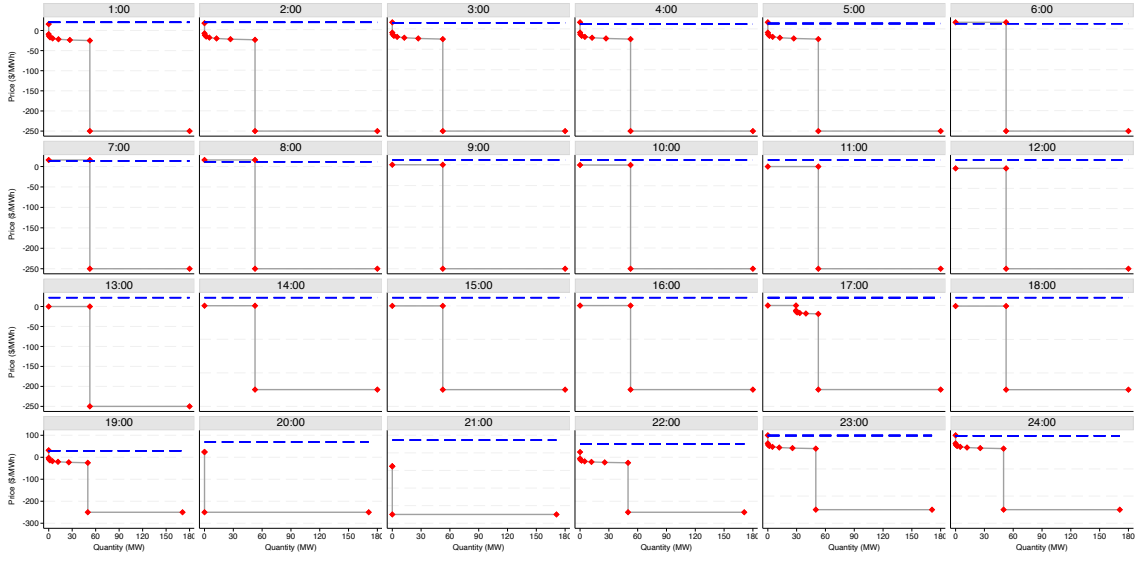
(b) censored bid price



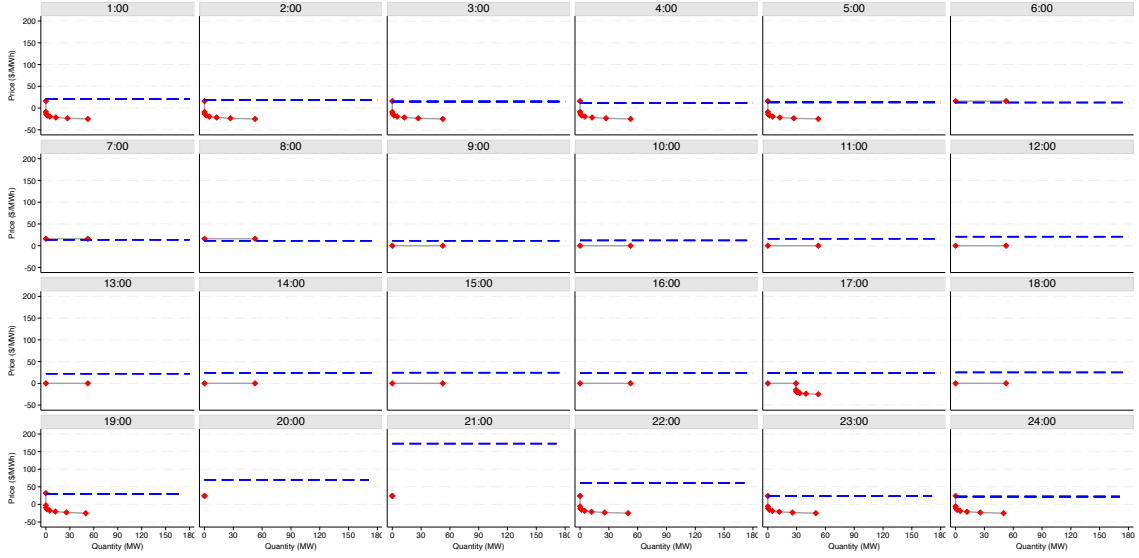
Notes: This figure plots the generation bids submitted by Turquoise Storage, one of the largest utility-scale battery in ERCOT. The bids are submitted on August 18, 2024 when Texas experiences significant heat, with temperatures reaching as high as 107°F in several locations. Panel (a) plots the raw bids and panel (b) plots the truncated bids at -10 dollars/MWh and 2,000 dollars/MWh. We use blue dashed line to denote the average real-time market prices for the corresponding hour in Texas on that day.

Figure A4: Load Bids of Turquoise Storage on August 18, 2024

(a) raw pattern

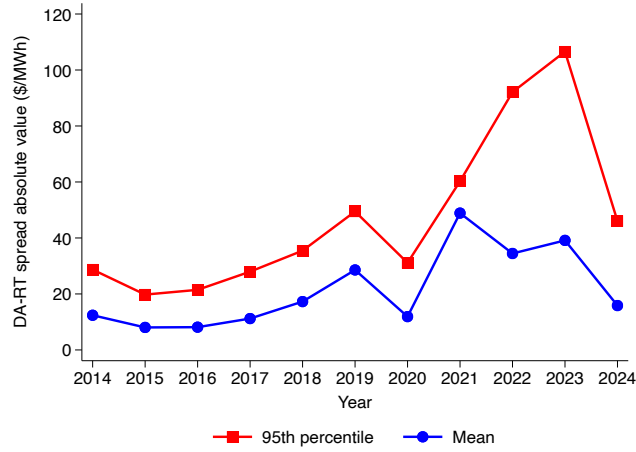


(b) censored bid price



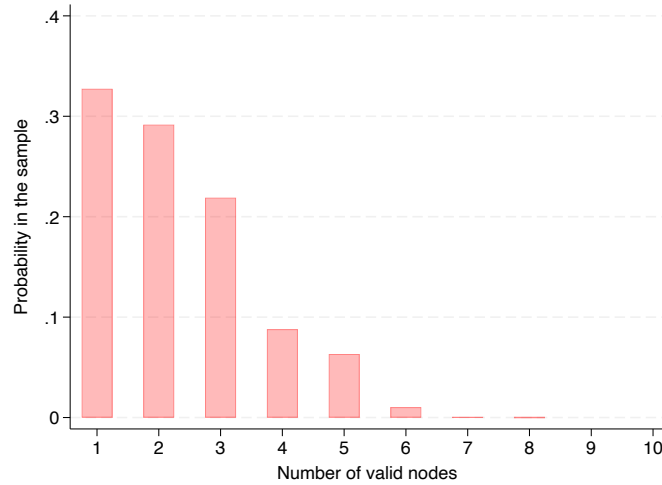
Notes: This figure plots the load bids submitted by Turquoise Storage, one of the largest utility-scale battery in ERCOT. The bids are submitted on August 18, 2024 when Texas experiences significant heat, with temperatures reaching as high as 107°F in several locations. Panel (a) plots the raw bids and panel (b) plots the truncated bids at -100 dollars/MWh and 1,000 dollars/MWh. We use blue dashed line to denote the average real-time market prices for the corresponding hour in Texas on that day.

Figure A5: DART Spread in ERCOT



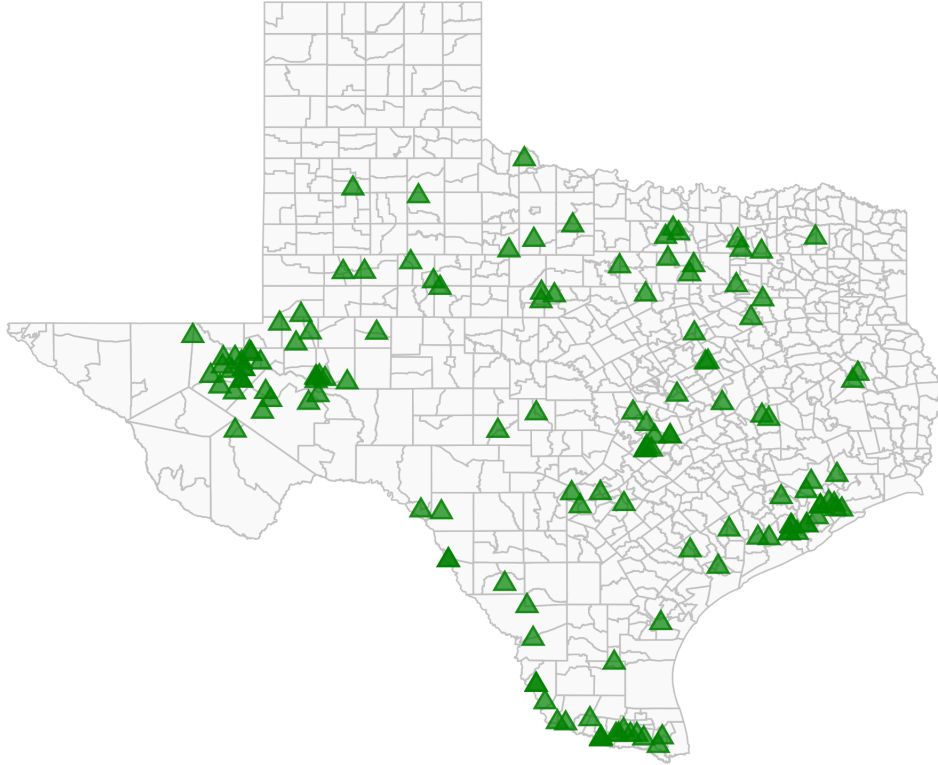
Notes: This figure plots the time series of absolute values of DART spread, which is defined as the difference between the day-ahead market price and real-time market price. We calculate the absolute values of DART spread at 15-minute frequency and calculate the annual averages (red circles) and annual 95th percentiles (blue triangles).

Figure A7: Numbers of Valid Bidding Segments



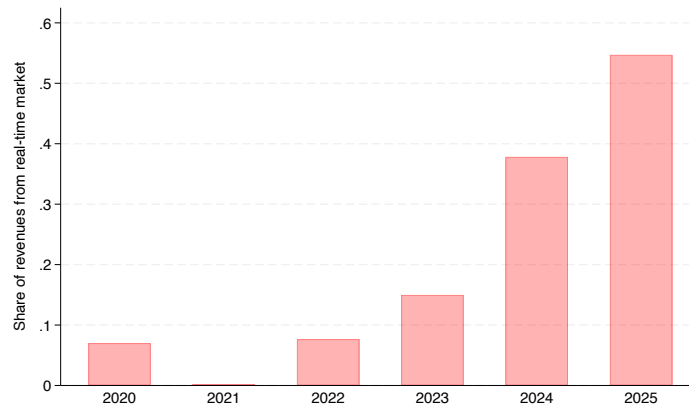
Notes: This figure plots the distribution of the numbers of valid bidding segments using bids information for each battery at 15-minute frequency in 2024. We define a bid segment as *valid* if its price lies between -\$50 and \$1,000/MWh, as the probability of prices falling outside this range is approximately 0.1% in either tail. In addition, we merge adjacent segments whose bid quantities differ by less than 0.1 MW or whose bid prices differ by less than \$0.1/MWh.

Figure A6: Map of Batteries in ERCOT



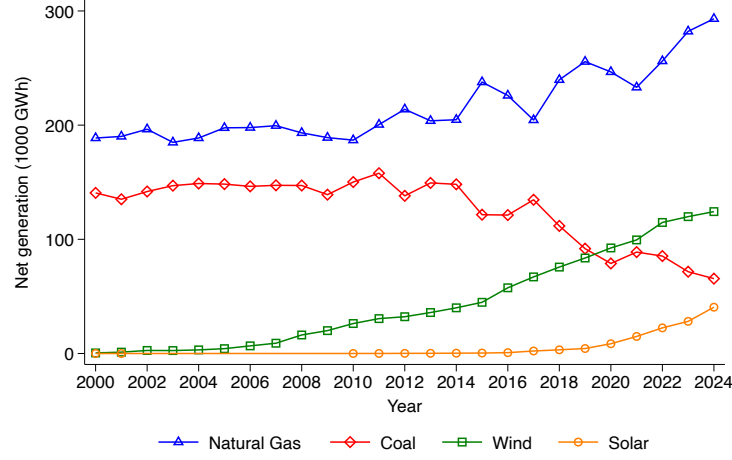
Notes: This figure plots the locations of utility-scale batteries in ERCOT in 2023. The coordinates of batteries are obtained from EIA-860.

Figure A8: Revenue Shares from Real-time Market



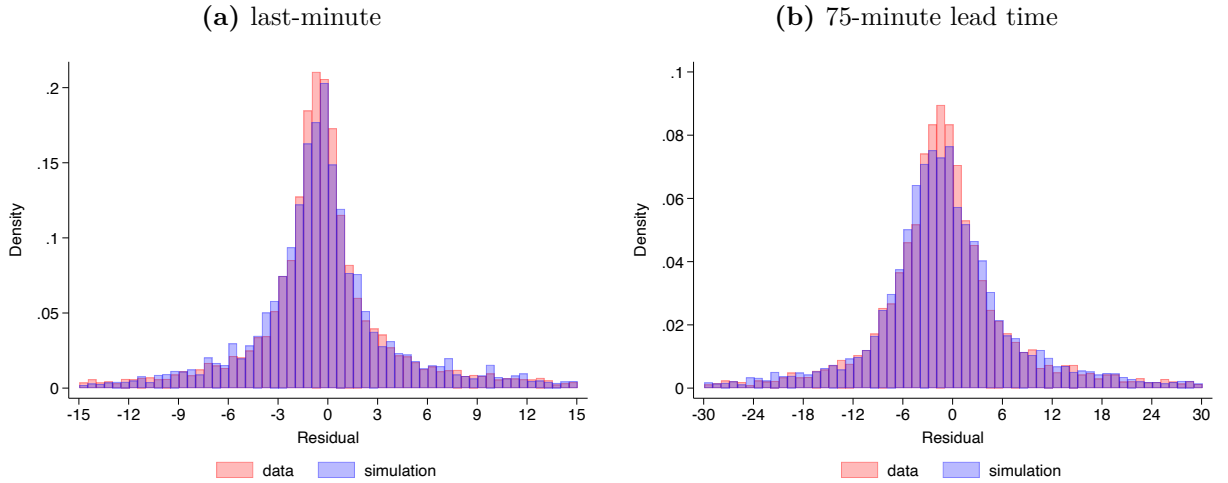
Notes: This figure plots the revenue shares from real-time market for batteries in ERCOT in 2020-2025. The revenue shares from real-time market are defined as the ratio of revenues from real-time energy arbitrage in total revenues from both real-time energy arbitrage and day-ahead ancillary service market.

Figure A9: Energy Composition in ERCOT



Notes: This figure plots the total electricity generation from different sources in ERCOT, compiled from EIA-923.

Figure A10: Price Forecast Errors: Data versus Model



Notes: This figure compares the forecast errors in price AR(1) models (equation (3)) between data and simulated prices. Panel (a) plots the distribution of residuals, when the price is lagged for 15 minutes, corresponding to the last-minute bidding case. Panel (b) plots the distribution of residuals, when the price is lagged for 90 minutes, corresponding to the 75-minute bid lead time case.

Table A2: Variation in Submitted Bid Quantities

	Bid Quantity (MW)			
	(1)	(2)	(3)	(4)
Adjusted R^2	0.794	0.866	0.812	0.023
Battery fixed effects	✓			
Battery-by-day fixed effects		✓		
Battery-by-interval fixed effects			✓	
Day-by-interval fixed effects				✓

Notes: This table explores the data variation in submitted bid quantities using bids information for each battery at 15-minute frequency in 2024. We control for battery fixed effects in column (1), battery-by-day fixed effects in column (2), battery-by-interval-of-day fixed effects in column (3), and day-by-interval fixed effects in column (4). We present the adjusted R^2 in the table.

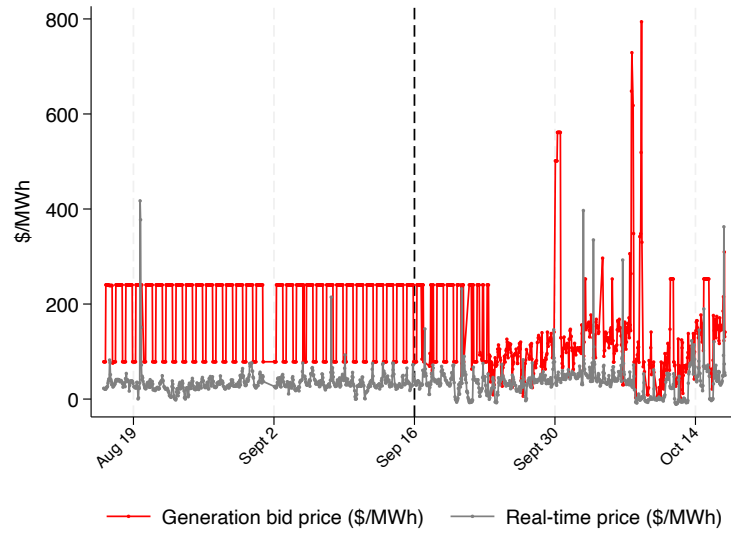
B Automated bidding adoption

We present a case study of automated bidding adoption in ERCOT, focusing on Flat Top (FLT), a one-hour battery with a nameplate capacity of 10 MW operated by Key Capture Energy.²² We focus on a sample spanning from August 16 to October 16, 2021. We begin by examining the evolution of average bid prices for FLT at the 15-minute level over this period. As shown in Figure A11, the variation in average bid prices increases substantially following the adoption of automated bidding.

We then compare hourly average bid prices and awarded quantities with the hourly pattern of real-time market prices, before and after September 16, 2021. Figure A12 shows that, prior to automation, bid prices take only two discrete values throughout the day, and awarded quantities are weakly aligned with real-time market prices. After automation, bid prices exhibit much richer intraday variation, and awarded quantities track real-time market prices more closely.

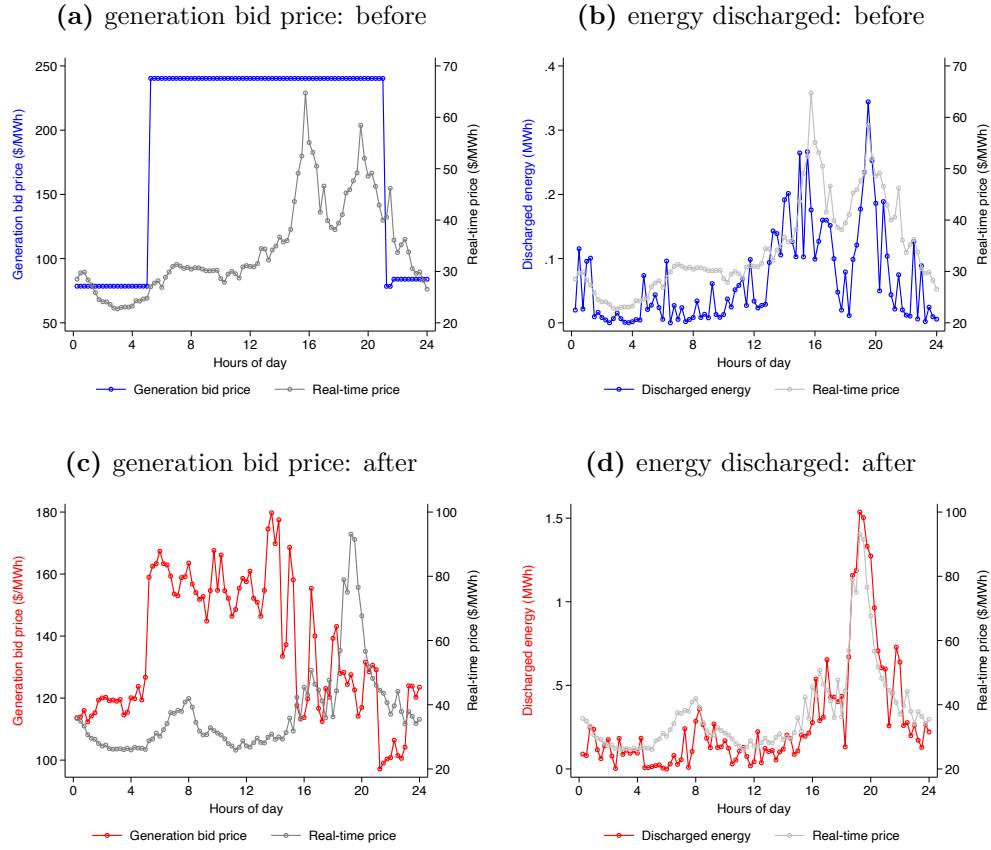
²²Another battery, Worsham (WRS), also a one-hour, 10 MW unit operated by the same firm, exhibits similar changes in bidding and operational patterns following the adoption of automated bidding on September 16, 2021. To streamline the presentation, we suppress the Worsham results in this draft. We thank Danny Musher of Key Capture Energy for sharing this institutional detail.

Figure A11: Automated Bidding Adoption by Flat Top: Average Bid Prices



Notes: This figure plots the 15-minute average generation bid prices and real-time electricity prices of Flat Top from August 16 to October 16, 2021. The vertical dashed line marks the date on which the battery adopted automated bidding (September 16, 2021).

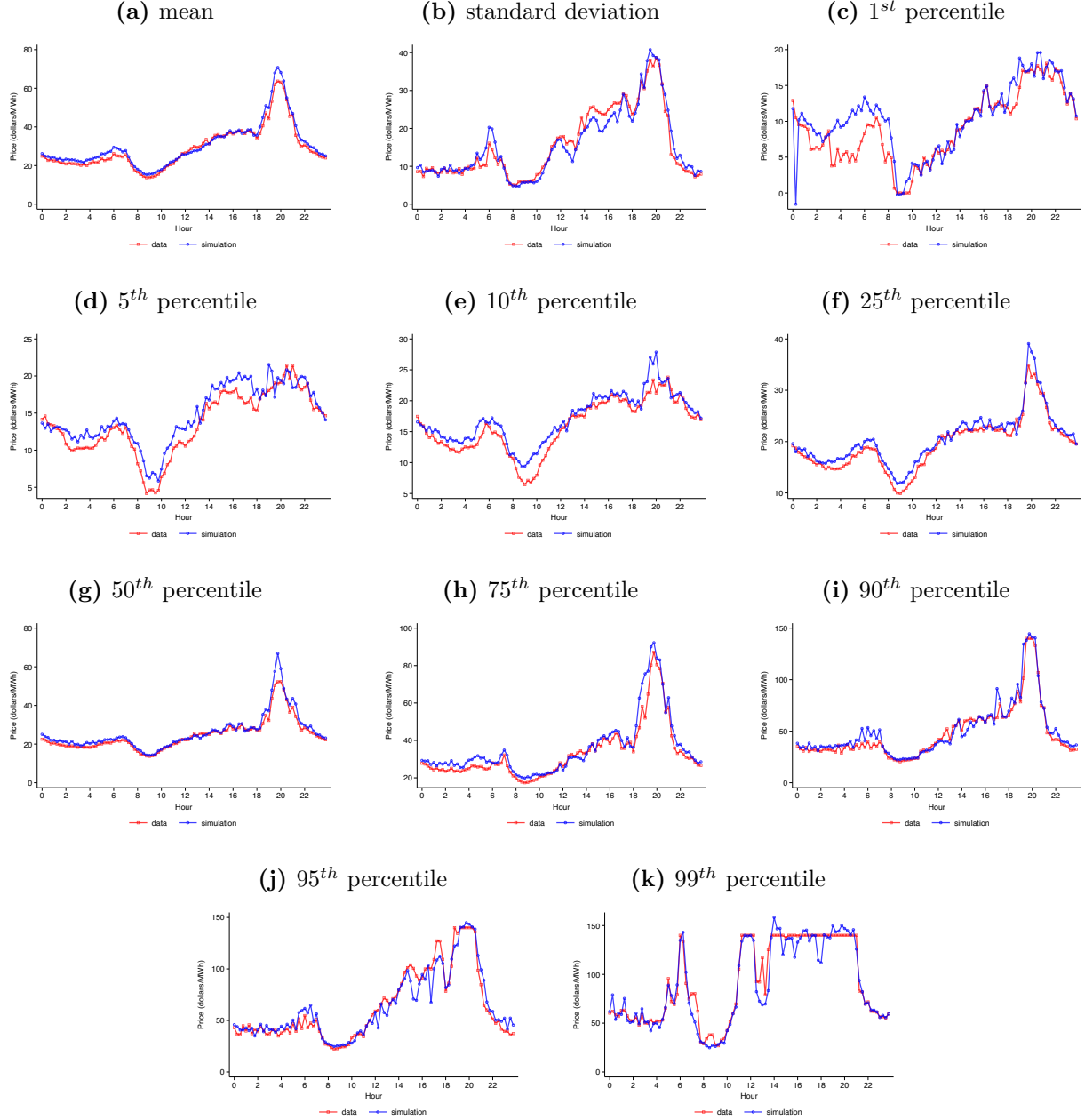
Figure A12: Automated Bidding Adoption by Flat Top: Intraday Patterns



Notes: This figure plots the bidding and operation of the battery Flat Top from August to October 2021, before and after it adopted automated bidding software. Panels (a) and (b) plot the generation bid prices and energy discharged by hour of day before the automated bidding adoption, and Panels (c) and (d) plot the generation bid prices and energy discharged by hour of day after the automated bidding adoption.

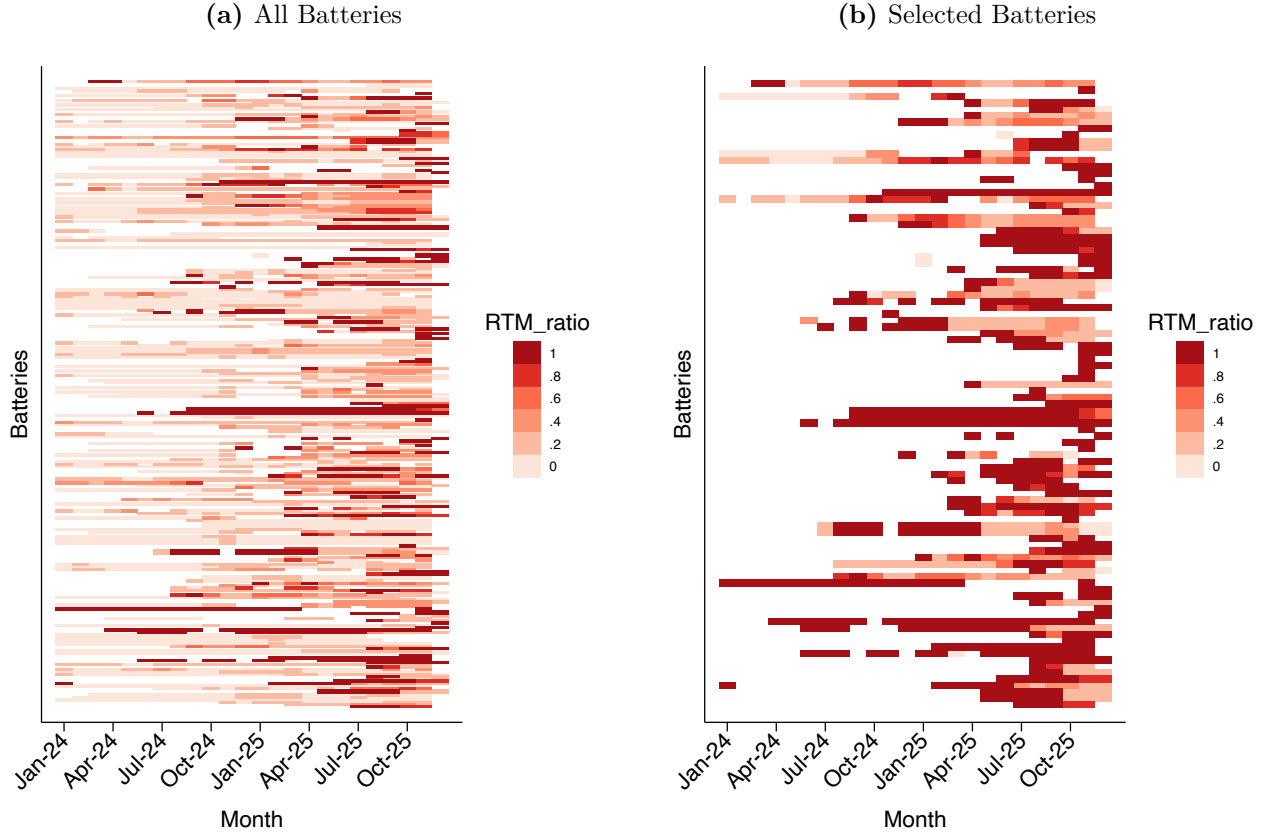
C Model fit and price-process diagnostics

Figure A13: Key Moments of Simulated Prices and Data



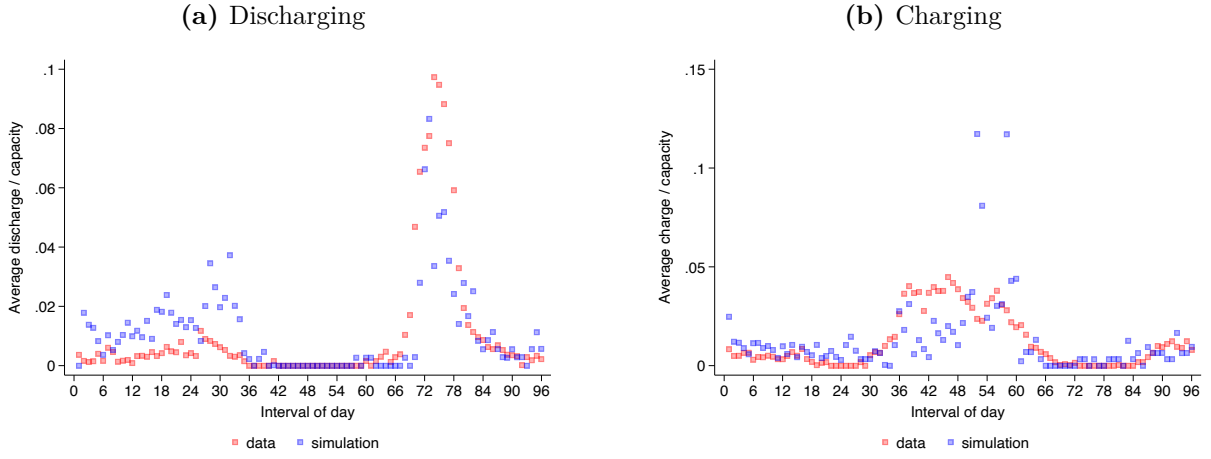
Notes: This figure compares the means (Panel (a)), standard deviations (Panel (b)), and a set of percentiles (Panel (c) to (k)) of simulated prices and realized prices at 15-minute frequency. The price data is from December 2024 to February 2025 at the settlement point where Casl Gap Battery sits. The statistics of simulated prices are denoted in red squares, and the statistics of realized price data are represented by blue circles.

Figure A14: Batteries Exclusive on RTM: March 2024 - November 2025



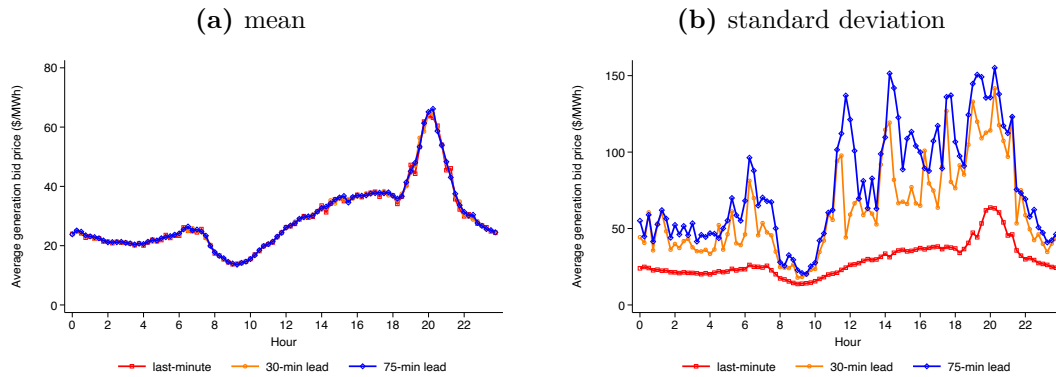
Notes: This figure plots the batteries' share of energy discharged into the real-time energy market in the total sum of both real-time energy market and day-ahead ancillary service market. Panel (a) plots the monthly shares for all active batteries in ERCOT from March 2024 to November 2025. Panel (b) plots the monthly shares for 39 batteries which has at least one season of exclusive real-time market operation between March 2024 and November 2025.

Figure A15: Comparison between model-predicted charging/discharging and data



Notes: This figure compares the model-predicted discharging pattern (Panel (a)) and charging pattern (Panel (b)) with data. The simulation is for Casl Gap in the summer of 2024.

Figure A16: Mean and Standard Deviations of Price Projections under Lead Time



Notes: This figure plots the means (Panel (a)) and standard deviations (Panel (b)) of projected prices under different lead times, conditional on the average prices in the starting interval. Red squares represent means and standard deviations of projected prices conditional on average prices 15 minutes ago (last-minute bidding). Orange circles represent means and standard deviations of projected prices conditional on average prices 45 minutes ago (with 30-minute lead time). Blue diamonds represent means and standard deviations of projected prices conditional on average prices 90 minutes ago (with 75-minute lead time).